

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384492
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Sakai; Masanori

Transmission device for human-powered vehicle and component for human-powered vehicle

Abstract

A transmission device for a human-powered vehicle basically includes a shaft member, a first planetary gear unit, and a second planetary gear unit. The first planetary gear unit includes a first planetary gear, a first carrier including a first carrier pin separate from a second carrier pin and arranged at a distance from the second carrier pin, a first sun gear, and a first ring gear. The second planetary gear unit includes a second planetary gear, a second carrier including the second carrier pin, a second sun gear, and a second ring gear. The first carrier pin and the second carrier pin are arranged overlapping a first plane. A rotational state of at least one of the first sun gear, the second sun gear, the first ring gear, and the second ring gear is controlled to switch the transmission stage in steps.

Inventors:	Sakai; Masanori (Osaka, JP)
Applicant:	Shimano Inc. (Osaka, JP)
Family ID:	95342565
Assignee:	Shimano Inc. (Osaka, JP)
Appl. No.:	18/769967
Filed:	July 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250136244 A1	May. 01, 2025

Foreign Application Priority Data

JP	2023-187130	Oct. 31, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: **B62M11/18** (20060101); **B62M11/14** (20060101); **F16H3/62** (20060101); **F16H3/66** (20060101); B62M6/55 (20100101)

U.S. Cl.:

CPC **B62M11/145** (20130101); **B62M11/18** (20130101); **F16H3/62** (20130101); **F16H3/666** (20130101); B62M6/55 (20130101)

Field of Classification Search

CPC: F16H (3/62); F16H (3/666); B62M (11/145); B62M (11/18)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2014/0121047	12/2013	Katsura	74/405	B62M 9/132

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
217294812	12/2021	CN	B62M 11/14
217301466	12/2021	CN	N/A
218258573	12/2022	CN	N/A
2018-100060	12/2017	JP	N/A

Primary Examiner: Lewis; Tisha D

Attorney, Agent or Firm: Global IP Counselors LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application No. 2023-187130, filed on Oct. 31, 2023. The entire disclosure of Japanese Patent Application No. 2023-187130 is hereby incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure generally relates to a transmission device for a human-powered vehicle and a component for a human-powered vehicle.

Background Information

(3) Japanese Laid-Open Patent Publication No. 2018-100060 (Patent Document 1) discloses an example of a component including a transmission device provided on a shaft member.

SUMMARY

(4) One objective of the present disclosure is to provide a transmission device for a human-powered vehicle and a component for a human-powered vehicle that are reduced in size in an axial direction of a shaft member.

(5) A transmission device in accordance with a first aspect of the present disclosure is for a human-powered vehicle. The transmission device is configured to switch a transmission stage of the human-powered vehicle in steps. The transmission device comprises a shaft member, a first planetary gear unit and a second planetary gear unit. The first planetary gear unit is provided on the shaft member. The second planetary gear unit is provided on the shaft member. The first planetary gear unit includes at least one first planetary gear, a first carrier, at least one first sun gear and at least one first ring gear. The first carrier includes at least one first carrier pin. The first carrier supports the at least one first planetary gear with the at least one first carrier pin. The at least one first sun gear is meshed with the at least one first planetary gear. The at least one first ring gear is meshed with the at least one first planetary gear. The second planetary gear unit includes at least one second planetary gear, a second carrier, at least one second planetary gear, at least one second sun gear, and at least one second ring gear. The second carrier includes at least one second carrier pin. The second carrier supports the at least one second planetary gear with the at least one second carrier pin. The at least one second sun gear is meshed with the at least one second planetary gear. The at least one second ring gear is meshed with the at least one second planetary gear. The at least one first carrier pin is a member separate from the at least one second carrier pin and arranged at a distance from the at least one second carrier pin. The at least one first carrier pin and the at least one second carrier pin are arranged overlapping a first plane that extends perpendicular to the shaft member. A rotational state of at least one of the at least one first sun gear, the at least one second sun gear, the at least one first ring gear, and the at least one second ring gear is controlled to switch the transmission stage in steps.

(6) In the transmission device according to the first aspect, the at least one first carrier pin and the at least one second carrier pin are arranged overlapping the first plane extending perpendicular to the shaft member. Thus, the transmission device is reduced in size in the axial direction.

(7) In accordance with a second aspect of the present disclosure, the transmission device according to the first aspect is configured so that the at least one first planetary gear includes a plurality of first planetary gears of different diameters. At least one of the first planetary gears having a maximum diameter is arranged overlapping the at least one second planetary gear as viewed in an axial direction of the shaft member.

(8) The transmission device according to the second aspect is reduced in size in the radial direction of the shaft member.

(9) In accordance with a third aspect of the present disclosure, the transmission device according to the first or second aspect is configured so that the at least one second planetary gear includes a plurality of second planetary gears of different diameters. At least one of the second planetary gears having a maximum diameter is arranged overlapping the at least one first planetary gear as viewed in an axial direction of the shaft member.

(10) The transmission device according to the third aspect is reduced in size in the radial direction of the shaft member.

(11) In accordance with a fourth aspect of the present disclosure, the transmission device according to any one of the first to third aspects further comprises a third planetary gear unit provided on the shaft member. The third planetary gear unit includes at least one third planetary gear, a third carrier, at least one third sun gear meshed with the at least one third planetary gear, and at least one third ring gear. The third carrier includes at least one third carrier pin. The third carrier supports the at least one third planetary gear with the at least one third carrier pin. The at least one third sun gear is meshed with the at least one third planetary gear. The at least one third ring gear is meshed with the at least one third planetary gear. The at least one third carrier pin is formed integrally with the at least one first carrier pin or the at least one second carrier pin. A rotational state of at least one of the at least one third sun gear and the at least one third ring gear is controlled to switch the transmission stage in steps.

(12) With the transmission device according to the fourth aspect, the rotational state of at least one

of the at least one first sun gear, the at least one second sun gear, the at least one first ring gear, the at least one second ring gear, the at least one third sun gear, and the at least one third ring gear is controlled to switch the transmission stage in steps in a preferred manner.

(13) In accordance with a fifth aspect of the present disclosure, the transmission device according to any one of the first to fourth aspects is configured so that part of the first carrier is formed integrally with part of the second carrier.

(14) With the transmission device according to the fifth aspect, part of the first carrier is formed integrally with part of the second carrier. Thus, the number of parts are reduced.

(15) In accordance with a sixth aspect of the present disclosure, the transmission device according to the fifth aspect is configured so that the first carrier includes a first carrier portion and a second carrier portion. The at least one first carrier pin is supported by the first carrier portion and the second carrier portion. The at least one first planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member.

(16) With the transmission device according to the sixth aspect, the at least one first carrier pin is supported by the first carrier portion and the second carrier portion. Thus, the at least one first carrier pin supports the at least one first planetary gear in a preferred manner.

(17) In accordance with a seventh aspect of the present disclosure, the transmission device according to the sixth aspect is configured so that the second carrier includes the first carrier portion and the second carrier portion. The at least one second carrier pin is supported by the first carrier portion and the second carrier portion. The at least one second planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member.

(18) With the transmission device according to the seventh aspect, the at least one second carrier pin is supported by the first carrier portion and the second carrier portion. Thus, the at least one second carrier pin supports the at least one second planetary gear in a preferred manner.

(19) In accordance with an eighth aspect of the present disclosure, the transmission device according to the fourth aspect is configured so that the first carrier includes a first carrier portion and a second carrier portion. The at least one first carrier pin is supported by the first carrier portion and the second carrier portion. The at least one first planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member. The second carrier includes a third carrier portion and at least one of the first carrier portion and the second carrier portion. The at least one second carrier pin is supported by the third carrier portion and the at least one of the first carrier portion and the second carrier portion. The at least one second planetary gear is located between the second carrier portion and the third carrier portion. The second carrier is located between the first carrier portion and the third carrier portion in an axial direction of the shaft member.

(20) With the transmission device according to the eighth aspect, the at least one first carrier pin is supported by the first carrier portion and the second carrier portion. Thus, the at least one first carrier pin supports the at least one first planetary gear in a preferred manner. With the transmission device according to the eighth aspect, the at least one second carrier pin is supported by the third carrier portion and at least one of the first carrier portion and the second carrier portion. Thus, the at least one second carrier pin supports the at least one second planetary gear in a preferred manner.

(21) In accordance with a ninth aspect of the present disclosure, the transmission device according to any one of the first to eighth aspects further comprises a transmission switching mechanism. The transmission switching mechanism is configured to switch the rotational state of the at least one first ring gear and the rotational state of the at least one second ring gear to switch the transmission stage in steps.

(22) With the transmission device according to the ninth aspect, the transmission switching mechanism switches the rotational state of the at least one first ring gear and the rotational state of the at least one second ring gear to switch the transmission stage in a preferred manner.

(23) In accordance with a tenth aspect of the present disclosure, the transmission device according to the ninth aspect is configured so that the at least one first planetary gear includes first planetary gears. The at least one first ring gear includes first ring gears that are rotatable independently from each other. The first planetary gears are each meshed with the first ring gears. The transmission switching mechanism is configured to switch rotational states of the first ring gears to switch the transmission stage in steps.

(24) With the transmission device according to the tenth aspect, the transmission switching mechanism switches rotational states of the first ring gears to switch the transmission stage in a preferred manner.

(25) In accordance with an eleventh aspect of the present disclosure, the transmission device according to the tenth aspect is configured so that the at least one second planetary gear includes second planetary gears. The at least one second ring gear includes second ring gears that are rotatable independently from each other. The second planetary gears are each meshed with the second ring gears. The transmission switching mechanism is configured to switch rotational states of the second ring gears to switch the transmission stage in steps.

(26) With the transmission device according to the eleventh aspect, the transmission switching mechanism switches rotational states of the second ring gears to switch the transmission stage in a preferred manner.

(27) In accordance with a twelfth aspect of the present disclosure, in the transmission device according to any one of the ninth to eleventh aspects, the at least one first sun gear and the at least one second sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

(28) With the transmission device according to the twelfth aspect, the at least one first sun gear and the at least one second sun gear are configured to rotate integrally in at least the first direction. Thus, driving force is transmitted in a preferred manner.

(29) In accordance with a thirteenth aspect of the present disclosure, the transmission device according to any one of the first to twelfth aspects is configured so that the at least one first planetary gear includes first planetary gears arranged at intervals about a rotational center axis of the shaft member. The at least one second planetary gear includes second planetary gears arranged at intervals about the rotational center axis of the shaft member. The at least one first carrier pin includes first carrier pins corresponding to the first planetary gears. The at least one second carrier pin includes second carrier pins corresponding to the second planetary gears. The first carrier pins and the second carrier pins are arranged alternately at equal intervals about the rotational center axis of the shaft member.

(30) With the transmission device according to the thirteenth aspect, the first planetary gears and the second planetary gears are arranged alternately at equal intervals about the rotational center axis of the shaft member. Thus, the first planetary gears and the second planetary gears are arranged stably.

(31) In accordance with a fourteenth aspect of the present disclosure, the transmission device according to any one of the first to thirteenth aspects further comprises a fourth planetary gear unit provided on the shaft member and a fifth planetary gear unit provided on the shaft member. The fourth planetary gear unit includes at least one fourth planetary gear, a fourth carrier, at least one fourth sun gear meshed with the at least one fourth planetary gear, and at least one fourth ring gear. The fourth carrier includes at least one fourth carrier pin. The fourth carrier supports the at least one fourth planetary gear with the at least one fourth carrier pin. The at least one fourth sun gear is meshed with the at least one fourth planetary gear. The at least one fourth ring gear is meshed with the at least one fourth planetary gear. The fifth planetary gear unit includes at least one fifth planetary gear, a fifth carrier, at least one fifth sun gear and at least one fifth ring gear. The fourth carrier includes at least one fifth carrier pin. The fifth carrier supports the at least one fifth planetary gear with the at least one fifth carrier pin. The at least one fifth sun gear is meshed with

the at least one fifth planetary gear. The at least one fifth ring gear is meshed with the at least one fifth planetary gear. The at least one fourth carrier pin is a member separate from the at least one fifth carrier pin and arranged at a distance from the at least one fifth carrier pin. The at least one fourth carrier pin and the at least one fifth carrier pin are arranged overlapping a second plane that extends perpendicular to the shaft member. A rotational state of at least one of the at least one fourth sun gear, the at least one fifth sun gear, the at least one fourth ring gear, and the at least one fifth ring gear is controlled to switch the transmission stage in steps.

(32) In the transmission device according to the fourteenth aspect, the at least one fourth carrier pin and the at least one fifth carrier pin are arranged overlapping the second plane extending perpendicular to the shaft member. Thus, the transmission device is reduced in size in the axial direction.

(33) In accordance with a fifteenth aspect of the present disclosure, the transmission device according to the fourteenth aspect is configured so that a part of the fourth carrier is formed integrally with a part of the fifth carrier.

(34) With the transmission device according to the fifteenth aspect, a part of the fourth carrier is formed integrally with a part of the fifth carrier. Thus, the number of parts are reduced.

(35) In accordance with a sixteenth aspect of the present disclosure, in the transmission device according to the fourteenth or fifteenth aspect, the at least one fourth sun gear and the at least one fifth sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

(36) With the transmission device according to the sixteenth aspect, the at least one fourth sun gear and the at least one fifth sun gear are configured to rotate integrally in at least the first direction. Thus, driving force is transmitted in a preferred manner.

(37) In accordance with a seventeenth aspect of the present disclosure, the transmission device according to any one of the fourteenth to sixteenth aspects further comprises an input rotational portion and an output rotational portion. The input rotational portion is connected to the first carrier. The output rotational portion is connected to the fifth carrier.

(38) With the transmission device according to the seventeenth aspect, rotational force that is input from the input rotational portion to the first carrier is transmitted from the fifth carrier to the output rotational portion.

(39) In accordance with an eighteenth aspect of the present disclosure, the transmission device according to any one of the fourteenth to seventeenth aspects is configured so that the at least one first sun gear, the at least one second sun gear, the at least one fourth sun gear, and the at least one fifth sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

(40) With the transmission device according to the eighteenth aspect, the at least one first sun gear, the at least one second sun gear, the at least one fourth sun gear, and the at least one fifth sun gear rotate integrally in at least the first direction. Thus, driving force is transmitted in a preferred manner.

(41) In accordance with a nineteenth aspect of the present disclosure, the transmission device according to any one of the first to thirteenth aspects further comprises a housing attachable to a frame of the human-powered vehicle. The transmission device is provided in the housing and configured to be connected to a rear wheel of the human-powered vehicle by at least one of a chain, a belt, and a shaft in a state in which the housing is attached to the frame.

(42) With the transmission device according to the nineteenth aspect, driving force that is output from the transmission device is transmitted to the rear wheel by at least one of a chain, a belt, and a shaft.

(43) In accordance with a twentieth aspect of the present disclosure, the transmission device according to any one of the first to nineteenth aspects further comprises an input rotational shaft. The input rotational shaft is arranged parallel to the shaft member and distanced from the shaft

member in a radial direction of the input rotational shaft.

(44) The transmission device according to the twentieth aspect is provided on the shaft member distanced from the input rotational shaft. Thus, the transmission device is reduced in size in the axial direction of the shaft member.

(45) A component in accordance with a twenty-first aspect of the present disclosure is for a human-powered vehicle. The component comprises the transmission device according to the first aspect, and a motor configured to apply a propulsion force to the human-powered vehicle.

(46) With the component according to the twenty-first aspect, the component including the motor configured to apply a propulsion force to the human-powered vehicle is reduced in size in the axial direction of the shaft member.

(47) The human-powered vehicle transmission device and the human-powered vehicle component according to the present disclosure are reduced in size in the axial direction of the shaft member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Referring now to the attached drawings which form a part of this original disclosure, an illustrative embodiment is shown.

(2) FIG. 1 is a side elevational view of a human-powered vehicle component including a human-powered vehicle transmission device in accordance with a first embodiment.

(3) FIG. 2 is a perspective view of the human-powered vehicle component shown in FIG. 1.

(4) FIG. 3 is a side view of the human-powered vehicle component shown in FIG. 1 without a housing in a first direction.

(5) FIG. 4 is a side view of the human-powered vehicle component shown in FIG. 1 without the housing in a second direction.

(6) FIG. 5 is a cross-sectional view of the human-powered vehicle component taken along section line D5-D5 in FIG. 3.

(7) FIG. 6 is an enlarged partial cross-sectional view of the human-powered vehicle component shown in FIG. 5.

(8) FIG. 7 is a front view of the human-powered vehicle transmission device shown in FIG. 1 without a ring gear.

(9) FIG. 8 is a perspective view of the human-powered vehicle transmission device shown in FIG. 1 without the ring gear.

(10) FIG. 9 is a perspective view of a first transmission unit shown in FIG. 7 without the ring gear.

(11) FIG. 10 is a plan view of the first transmission unit shown in FIG. 7 without the ring gear.

(12) FIG. 11 is a perspective view of a second transmission unit shown in FIG. 7 without the ring gear.

(13) FIG. 12 is a plan view of the second transmission unit shown in FIG. 7 without the ring gear.

(14) FIG. 13 is a cross-sectional view of a part of the human-powered vehicle transmission device shown in FIG. 1 and a part of the human-powered vehicle transmission device shown in FIG. 6 without the ring gear.

(15) FIG. 14 is a cross-sectional view of a part of the human-powered vehicle transmission device shown in FIG. 1 and the human-powered vehicle transmission device without the ring gear taken along section line D14-D14 in FIG. 3.

(16) FIG. 15 is a cross-sectional view of the transmission device taken along section line D15-D15 in FIG. 6.

(17) FIG. 16 is a cross-sectional view of the transmission device taken along section line D16-D16 in FIG. 6.

(18) FIG. 17 is a skeleton diagram showing a transmission path of a driving force of the human-

powered vehicle component shown in FIG. 6.

(19) FIG. 18 is a perspective view of a transmission switching mechanism of the human-powered vehicle transmission device shown in FIG. 6.

(20) FIG. 19 is a side view showing a human-powered vehicle component without a housing in accordance with a second embodiment in the first direction.

(21) FIG. 20 is a side view showing the human-powered vehicle component without the housing in accordance with the second embodiment in the second direction.

(22) FIG. 21 is a skeleton diagram showing a transmission path of a driving force of a human-powered vehicle component in accordance with the second embodiment.

DETAILED DESCRIPTION

(23) Selected embodiments will now be explained with reference to the drawings. It will be apparent to those skilled in the bicycle field from this disclosure that the following descriptions of the embodiments are provided for illustration only and not for the purpose of limiting the invention as defined by the appended claims and their equivalents.

First Embodiment

(24) A first embodiment of a transmission device 22 for a human-powered vehicle and a component 20 for a human-powered vehicle will now be described with reference to FIGS. 1 to 18.

(25) The human-powered vehicle is a vehicle including at least one wheel and driven by at least a human driving force. The human-powered vehicle includes, for example, various types of bicycles such as a mountain bike, a road bike, a city bike, a cargo bike, a hand bike, and a recumbent bike. The number of wheels on the human-powered vehicle is not limited. The human-powered vehicle includes, for example, a unicycle and a vehicle including two or more wheels. The human-powered vehicle is not limited to a vehicle configured to be driven only by a human driving force. The human-powered vehicle includes an E-bike that uses a driving force of an electric motor in addition to a human driving force for propulsion. The E-bike includes an electric assist bicycle that assists in propulsion with an electric motor. In the embodiments described below, the human-powered vehicle refers to a bicycle.

(26) The human-powered vehicle includes, for example, a vehicle body and a wheel. The vehicle body includes, for example, a frame. The wheel is, for example, provided on the frame. The wheel includes a front wheel and a rear wheel. The human-powered vehicle includes a crank axle 10 rotatable relative to the frame. A crank arm 12 is provided on each axial end of the crank axle 10. A pedal is coupled to each crank arm 12. The human driving force is input to the crank axle 10 from the pedals and the crank arms 12.

(27) The component 20 is, for example, provided on a transmission path of the human driving force between the crank axle 10 and a driving wheel of the human-powered vehicle. The driving wheel includes, for example, at least one of the front wheel and the rear wheel. The driving wheel includes, for example, the rear wheel. The component 20 is disposed around an input rotational center axis C1 of the crank axle 10. The component 20 can be provided on an axle of the driving wheel.

(28) The component 20 for a human-powered vehicle includes, for example, a transmission device 22, a shaft member 24 distanced from an input rotational shaft 28 of the human-powered vehicle, and a speed increaser 26. The transmission device 22 is arranged on the shaft member 24.

(29) The transmission device 22 is, for example, configured to shift a final transmission ratio of the human-powered vehicle. The final transmission ratio of the human-powered vehicle refers to, for example, a ratio of rotational speed of the driving wheel to rotational speed of the crank axle 10. The final transmission ratio of a human-powered vehicle is determined based on, for example, the transmission ratio of the transmission device included in the transmission path of the human driving force, the reduction ratio of a speed reducer included in the transmission path of the human driving force, and the increase ratio of the speed increaser included in the transmission path of the human driving force.

- (30) The transmission device **22** is, for example, configured to shift the transmission ratio. The transmission device **22** is, for example, configured to select one of transmission ratios. The transmission ratio refers to, for example, a ratio of the output rotational speed from the transmission device **22** to the input rotational speed of the transmission device **22**. The transmission ratio can be greater than one and less than one. The transmission ratio can be one.
- (31) In this specification, the transmission ratio, the reduction ratio, and the increase ratio are each expressed by a speed ratio R that is a ratio of the output rotational speed $S2$ to the input rotational speed $S1$ input to the corresponding device. The speed ratio R is expressed by Equation 1.
- $R=S2/S1$ Equation 1:
- (32) In a case where $S1=S2$, the speed ratio R is one. In a case where $S1<S2$, the speed ratio R is greater than one. In a case where $S1>S2$, the speed ratio R is less than one. The speed-increasing rate increases, for example, as the increase ratio becomes greater. The speed-decreasing rate decreases, for example, as the reduction ratio becomes greater.
- (33) The component **20** further includes, for example, the input rotational shaft **28**. In an example, the human driving force is input to the input rotational shaft **28**. The input rotational shaft **28** forms, for example, part of the transmission path of the human driving force. The input rotational shaft **28** includes, for example, the crank axle **10**. The shaft member **24** is, for example, substantially parallel to the input rotational shaft **28**. In this specification, the phrase of “substantially parallel” refers to, for example, being completely parallel and being considered to be completely parallel. The case of being considered to be completely parallel refers to, for example, a case where the function of the component **20** is not significantly affected by the non-parallel state.
- (34) The speed increaser **26** is, for example, provided in the transmission path of the human driving force. The speed increaser **26** forms, for example, part of the transmission path of the human driving force. The speed increaser **26** is, for example, connected to the input rotational shaft **28**. The speed increaser **26** is, for example, configured to receive the human driving force through the input rotational shaft **28**. The speed increaser **26** is, for example, configured to transmit the human driving force input to the input rotational shaft **28** to the transmission device **22**. The speed increaser **26** is, for example, disposed to rotate about the rotational center axis $C2$ of the shaft member **24**. The speed increaser **26** can be disposed to rotate about the input rotational center axis $C1$ of the input rotational shaft **28**.
- (35) The transmission device **22** is, for example, provided in the transmission path of the human driving force. The transmission device **22** forms, for example, part of the transmission path of the human driving force. The transmission device **22** is, for example, connected to the speed increaser **26**. The human driving force input to the input rotational shaft **28** is input to the transmission device **22** through the speed increaser **26**. The transmission device **22** is, for example, disposed to rotate about the rotational center axis $C2$ of the shaft member **24**.
- (36) The component **20** further includes, for example, a housing **30** and an output rotational portion **32**. The output rotational portion **32** is, for example, provided on the transmission path of the human driving force. The output rotational portion **32** forms, for example, part of the transmission path of the human driving force. The output rotational portion **32** is, for example, disposed coaxially with the shaft member **24**. The output rotational portion **32** is, for example, connected to the transmission device **22**.
- (37) In an example, driving force output from the transmission device **22** is transmitted to the output rotational portion **32**. The output rotational portion **32** is, for example, configured to output the human driving force to the driving wheel. The output rotational portion **32** is, for example, configured to transmit the driving force to the rear wheel through a linking member **14**.
- (38) The component **20** further includes, for example, an output rotor **34**. The output rotor **34** is, for example, configured to output the driving force that is output from the transmission device **22**. The output rotor **34** is, for example, provided on the transmission path of the human driving force. The output rotor **34** forms, for example, part of the transmission path of the human driving force. The

output rotor **34** is, for example, disposed coaxially with the shaft member **24**. The output rotor **34** is, for example, connected to the output rotational portion **32**.

(39) The output rotor **34** is, for example, connected to the wheel by the linking member **14**. The output rotational portion **32** is, for example, configured to transmit the driving force to the rear wheel through the output rotor **34** and the linking member **14**. The output rotational portion **32** includes, for example, a sprocket mount **32A**. The output rotor **34** includes, for example, one of a sprocket, a pulley, and a bevel gear. The linking member **14** includes, for example, at least one of a chain, a belt, and a shaft. In an example, a sprocket is mounted on the sprocket mount **32A**. The sprocket mount **32A** is, for example, configured to transmit driving force to the rear wheel through the sprocket and the chain.

(40) The rotational direction of the output rotor **34** is, for example, substantially equal to the rotational direction of the input rotational shaft **28**. The speed increaser **26** and the transmission device **22** are, for example, configured to rotate the output rotor **34** forward in a case where the input rotational shaft **28** rotates forward. The speed increaser **26** and the transmission device **22** can be configured to, for example, rotate the output rotor **34** rearward in a case where the input rotational shaft **28** rotates rearward. A first one-way clutch is provided on the transmission path of the human driving force between the input rotational shaft **28** and the output rotor **34**. The first one-way clutch includes at least one of a roller clutch, a pawl-type clutch, and a sprag-type clutch. The first one-way clutch is, for example, provided on at least one of the speed increaser **26** and the transmission device **22**. The first one-way clutch is, for example, configured to allow rotation of the input rotational shaft **28** relative to the output rotor **34** in a case where the input rotational shaft **28** rotates rearward. The forward rotational direction of the input rotational shaft **28** corresponds to, for example, the rotational direction of the input rotational shaft **28** in a case where the human-powered vehicle moves forward. The forward rotational direction of the output rotor **34** corresponds to, for example, the rotational direction of the output rotor **34** in a case where the human-powered vehicle moves forward.

(41) Preferably, a second one-way clutch is provided in the transmission path of human driving force between the output rotational portion **32** and the rear wheel. In an example, the second one-way clutch rotates the driving wheel forward in a case where the output rotor **34** rotates forward. The second one-way clutch is, for example, configured to restrict rearward rotation of the driving wheel in a case where the output rotor **34** rotates rearward. The second one-way clutch includes, for example, at least one of a roller clutch, a pawl-type clutch, and a sprag-type clutch.

(42) The housing **30** is, for example, configured to be mounted on the frame of the human-powered vehicle. The housing **30** includes, for example, at least one frame coupling portion **30A**. The at least one frame coupling portion **30A** is coupled to the frame of the human-powered vehicle by, for example, a bolt. The at least one frame coupling portion **30A** includes, for example, a plurality of frame coupling portions **30A** spaced apart from each other about the input rotational center axis **C1**. The frame coupling portions **30A** are separated from one another by 90 degrees or greater about the input rotational center axis **C1**. The housing **30** is, for example, configured to support the input rotational shaft **28**. The input rotational shaft **28** is, for example, provided on the housing **30** and is rotatable relative to the housing **30** about the input rotational center axis **C1**.

(43) In an example, at least part of a driving force transmission member of the component **20** is provided on the housing **30**. The driving force transmission member includes, for example, at least one of the transmission device **22**, the shaft member **24**, the speed increaser **26**, the input rotational shaft **28**, the output rotational portion **32**, and the output rotor **34**. In an example, the input rotational shaft **28**, the speed increaser **26**, and the transmission device **22** are accommodated in the housing **30**. In an example, the shaft member **24** and the output rotational portion **32** are provided on the housing **30**.

(44) The component **20** includes, for example, an inner cavity **20A** surrounded by the housing **30**. The transmission device **22**, the shaft member **24**, and the speed increaser **26** are, for example,

disposed in the inner cavity **20A**. At least part of the input rotational shaft **28** and at least part of the output rotational portion **32** are, for example, disposed in the inner cavity **20A** of the housing **30**.

(45) At least one of the transmission device **22**, the speed increaser **26**, the output rotational portion **32**, and the output rotor **34** includes, for example, a part formed from a metal material. At least one of the transmission device **22**, the speed increaser **26**, the output rotational portion **32**, and the output rotor **34** can include a part formed from a resin material. At least one of the transmission device **22**, the speed increaser **26**, the output rotational portion **32**, and the output rotor **34** includes, for example, a part that forms the transmission path of human driving force and is at least partially formed from a metal material. At least one of the transmission device **22**, the speed increaser **26**, the output rotational portion **32**, and the output rotor **34** can include a part that forms the transmission path of human driving force and is at least partially formed from a resin material.

(46) The speed increaser **26** includes, for example, at least one first planetary gear group **36**. The at least one first planetary gear group **36** is, for example, rotatable about the rotational center axis **C2** of the shaft member **24**. The at least one first planetary gear group **36**, the transmission device **22**, and the output rotor **34** are, for example, arranged in an axial direction **A1** of the shaft member **24** in an order of the at least one first planetary gear group **36**, the transmission device **22**, and the output rotor **34**.

(47) The at least one first planetary gear group **36** includes, for example, at least one speed-increasing planetary gear **36A**, a speed-increasing carrier **36B**, at least one speed-increasing sun gear **36C**, and at least one speed-increasing ring gear **36D**. The at least one speed-increasing planetary gear **36A** is, for example, disposed between the at least one speed-increasing sun gear **36C** and the at least one speed-increasing ring gear **36D** in a radial direction with respect to the rotational center axis **C2**. The at least one speed-increasing sun gear **36C** is, for example, meshed with the at least one speed-increasing planetary gear **36A**. The at least one speed-increasing ring gear **36D** is, for example, meshed with the at least one speed-increasing planetary gear **36A**.

(48) The speed-increasing carrier **36B** includes at least one speed-increasing carrier pin **36E**. The speed-increasing carrier **36B**, for example, supports the at least one speed-increasing planetary gear **36A** with the at least one speed-increasing carrier pin **36E**. The at least one speed-increasing carrier pin **36E** includes two ends in the axial direction **A1** defining a first speed-increasing pin end **36F** and a second speed-increasing pin end **36G**. The speed-increasing carrier **36B** is, for example, supported on the housing **30** so that rotation of the speed-increasing carrier **36B** relative to the housing **30** is restricted. The restriction of rotation of the speed-increasing carrier **36B** relative to the housing **30** restricts revolution of the speed-increasing planetary gear **36A** around the speed-increasing sun gear **36C**.

(49) The speed-increasing carrier **36B** includes, for example, a first speed-increasing carrier portion **36H** and a second speed-increasing carrier portion **36K**. The first speed-increasing pin end **36F** is, for example, disposed on the first speed-increasing carrier portion **36H**. The second speed-increasing pin end **36G** is, for example, disposed on the second speed-increasing carrier portion **36K**. The at least one speed-increasing planetary gear **36A** is, for example, disposed between the first speed-increasing carrier portion **36H** and the second speed-increasing carrier portion **36K** in the axial direction **A1**.

(50) The speed increaser **26** further includes, for example, a speed-increasing unit **38** that differs from the at least one first planetary gear group **36**. The speed-increasing unit **38** is, for example, configured to transmit the human driving force that is input to the input rotational shaft **28** to the at least one first planetary gear group **36**.

(51) The speed-increasing unit **38** includes, for example, a first speed-increasing gear **40** arranged coaxially with the input rotational shaft **28** and a second speed-increasing gear **42** meshed with the first speed-increasing gear **40** and arranged coaxially with the shaft member **24**. The first speed-increasing gear **40** includes, for example, a spur gear. The second speed-increasing gear **42** includes, for example, a spur gear. The first speed-increasing gear **40**, for example, rotates

integrally with the input rotational shaft **28**. The first speed-increasing gear **40** has, for example, a larger pitch circle diameter than the second speed-increasing gear **42**. The second speed-increasing gear **42** and the input rotational shaft **28**, for example, rotate in opposite directions. The at least one first planetary gear group **36** is, for example, located between the second speed-increasing gear **42** and the transmission device **22** in the axial direction **A1** of the shaft member **24**. The first speed-increasing gear **40** and the second speed-increasing gear **42** can each include a helical gear.

(52) The speed-increasing unit **38** can include sprockets and a chain engaging the sprockets instead of or in addition to the first speed-increasing gear **40** and the second speed-increasing gear **42**. The speed-increasing unit **38** can include pulleys and a belt engaging the pulleys instead of or in addition to the first speed-increasing gear **40** and the second speed-increasing gear **42**.

(53) The at least one first planetary gear group **36** includes, for example, a speed-increasing input portion **44** and a speed-increasing output portion **46**. The speed-increasing input portion **44** is, for example, formed integrally with the second speed-increasing gear **42**. The speed-increasing input portion **44** and the second speed-increasing gear **42** can be molded integrally so as to be a single-piece member. In a case where the speed-increasing input portion **44** and the second speed-increasing gear **42** are molded integrally, the assembling process will be simplified. The second speed-increasing gear **42** is, for example, provided on an outer circumference of the speed-increasing input portion **44**. The rotational force is increased in speed by the speed-increasing unit **38** and is input to the speed-increasing input portion **44**. The speed increaser **26** has a speed increasing ratio that is, for example, a ratio of the rotational speed of the speed-increasing output portion **46** to the rotational speed of the first speed-increasing gear **40**.

(54) The speed-increasing input portion **44** includes, for example, a speed-increasing ring gear **36D**. The speed-increasing ring gear **36D** is, for example, provided on an inner circumference of the speed-increasing input portion **44**. The speed-increasing ring gear **36D** is, for example, provided on a part of the speed-increasing input portion **44** differing from the second speed-increasing gear **42** in the axial direction **A1**. The speed-increasing ring gear **36D** has, for example, a larger pitch circle diameter than the second speed-increasing gear **42**. The speed-increasing output portion **46** includes, for example, the speed-increasing sun gear **36C**.

(55) The speed-increasing input portion **44** and the speed-increasing output portion **46** are, for example, configured to rotate in opposite directions. For example, the rotation of the speed-increasing carrier **36B** relative to the housing **30** is restricted. Thus, the rotational force that is input to the speed-increasing ring gear **36D** is output from the speed-increasing sun gear **36C** as a rotational force in the opposite direction. The rotational direction of the speed-increasing input portion **44** is, for example, the same as the rotational direction of the second speed-increasing gear **42**. Since the second speed-increasing gear **42** and the input rotational shaft **28** rotate in opposite directions, the speed-increasing input portion **44** and the input rotational shaft **28**, for example, rotate in opposite directions. Thus, the rotational direction of the speed-increasing output portion **46** about the rotational center axis **C2** of the shaft member **24** is the same as the rotational direction of the input rotational shaft **28** about the input rotational center axis **C1**.

(56) In an example, the speed-increasing ratio of the speed increaser **26** is 3 or greater and 20 or less. In an example, the speed-increasing ratio of the speed increaser **26** is 5 or greater and 18 or less. In an example, the speed-increasing ratio of the speed increaser **26** is 7 or greater and 16 or less. In an example, the speed-increasing ratio of the speed increaser **26** is 9 or greater and 14 or less. In an example, the speed-increasing ratio of the speed increaser **26** is 9.4. The speed-increasing ratio of the speed increaser **26** can be 3 or greater and 11 or less. The speed-increasing ratio of the speed increaser **26** can be 6 or greater and 20 or less. In an example, the speed-increasing ratio of the speed increaser **26** is 6.032.

(57) The transmission device **22** for a human-powered vehicle is, for example, configured to switch a transmission stage of the human-powered vehicle in steps. In an example, the transmission device **22** includes a plurality of transmission stages. In an example, the transmission stages each have a

different transmission ratio. The transmission stage is changed from one transmission stage to another transmission stage to shift the transmission ratio of the transmission device **22**.

(58) The transmission device **22** includes, for example, at least one secondary planetary gear group **48**. The transmission device **22** includes, for example, the shaft member **24**, a first planetary gear unit **50** provided on the shaft member **24**, and a second planetary gear unit **52** provided on the shaft member **24**. The at least one secondary planetary gear group **48** includes, for example, the first planetary gear unit **50** and the second planetary gear unit **52**.

(59) The at least one secondary planetary gear group **48** includes, for example, a plurality of transmission planetary gears **48A**. The at least one secondary planetary gear group **48** includes, for example, a plurality of transmission carriers **48B**. The transmission carriers **48B** support, for example, the transmission planetary gears **48A**. The at least one secondary planetary gear group **48** includes, for example, a plurality of transmission sun gears **48C**. The transmission sun gears **48C** are, for example, meshed with the transmission planetary gears **48A**. The at least one secondary planetary gear group **48** includes, for example, a plurality of transmission ring gears **48D**. The transmission ring gears **48D** are, for example, meshed with the transmission planetary gears **48A**.

(60) The transmission device **22** further includes, for example, the input rotational shaft **28**. The input rotational shaft **28** is, for example, disposed parallel to the shaft member **24**, and spaced from the shaft member **24** in a radial direction of the input rotational shaft **28**.

(61) The transmission device **22** further includes, for example, the housing **30** attachable to the frame of the human-powered vehicle. In an example, the transmission device **22** is provided on the housing **30**. The transmission device **22** has a transmission rotational center axis that is, for example, substantially the same as the rotational center axis **C2** of the shaft member **24**.

(62) The transmission device **22** further includes, for example, an input rotational portion **54** and the output rotational portion **32**. The input rotational portion **54** is, for example, configured to transmit a rotational force from the speed-increasing output portion **46** to the transmission device **22**. The input rotational portion **54** includes, for example, a first input rotational portion **54A** and a second input rotational portion **54B**. The first input rotational portion **54A** and the second input rotational portion **54B** include, for example, gears that are meshed with each other. The first input rotational portion **54A** is, for example, formed integrally with the speed-increasing output portion **46**. The second input rotational portion **54B** is, for example, formed integrally with a first carrier portion **72**.

(63) The transmission device **22** is, for example, configured so that the ratio of the rotational speed of the output rotational portion **32** to the rotational speed of the input rotational portion **54** corresponds to the transmission ratio of the transmission device **22**. The at least one secondary planetary gear group **48** is, for example, configured to output the rotational force to the output rotational portion **32**. The transmission device **22** is, for example, configured to be connected to the rear wheel of the human-powered vehicle by at least one of a chain, a belt, and a shaft in a state in which the housing **30** is attached to the frame.

(64) The transmission ratio of the transmission device **22** is, for example, a ratio of the rotational speed of the output rotational portion **32** to the rotational speed of the input rotational portion **54**. In an example, the transmission device **22** is configured to shift the transmission ratio within a range from a minimum transmission ratio to a maximum transmission ratio. The minimum transmission ratio is, for example, a transmission ratio corresponding to a speed decrease. The maximum transmission ratio is, for example, a transmission ratio corresponding to a speed increase. In an example, the minimum transmission ratio is greater than 0 and less than 1. In an example, the maximum transmission ratio is greater than 1 and less than or equal to 10. As the minimum transmission ratio decreases, the speed-decreasing rate becomes larger in a case where the transmission ratio is the minimum transmission ratio. As the maximum transmission ratio increases, the speed-increasing rate becomes larger in a case where the transmission ratio is the maximum transmission ratio.

(65) The transmission stages include, for example, a minimum transmission stage and a maximum transmission stage. The transmission ratio of the minimum transmission stage is, for example, the minimum transmission ratio. The transmission ratio of the maximum transmission stage is, for example, the maximum transmission ratio. The ratio of the maximum transmission ratio to the minimum transmission ratio is, for example, 1.2 or greater and 8 or less. In an example, the ratio of the maximum transmission ratio to the minimum transmission ratio is 1.7 or greater and 7 or less. In an example, the ratio of the maximum transmission ratio to the minimum transmission ratio is 2.4 or greater and 6 or less. In an example, the ratio of the maximum transmission ratio to the minimum transmission ratio is 2.4.

(66) The at least one secondary planetary gear group **48** is, for example, configured to shift the transmission ratio in three or more steps. In an example, the transmission stages include three or more transmission stages. In an example, the three or more transmission stages each have a different transmission ratio.

(67) The first planetary gear unit **50** includes, for example, at least one first planetary gear **56**, a first carrier **58**, at least one first sun gear **60**, and at least one first ring gear **62**.

(68) The second planetary gear unit **52** includes, for example, at least one second planetary gear **64**, a second carrier **66**, at least one second sun gear **68**, and at least one second ring gear **70**.

(69) The transmission planetary gears **48A** include, for example, the at least one first planetary gear **56**. The at least one first planetary gear **56** includes, for example, a plurality of first planetary gears **56**. The at least one first planetary gear **56** includes, for example, a plurality of the first planetary gears **56** arranged at intervals about the rotational center axis **C2** of the shaft member **24**. The first planetary gears **56** are arranged at intervals about the rotational center axis **C2** of the shaft member **24**. The number of the first planetary gears **56** is, for example, two or greater and eight or less. In the present embodiment, the number of the first planetary gears **56** is four.

(70) The at least one first planetary gear **56** includes, for example, a plurality of first planetary gears **56** of different diameters. The first planetary gears **56** include a first transmission planetary gear **56A** and a second transmission planetary gear **56B** having a larger pitch circle diameter than the first transmission planetary gear **56A**. The first transmission planetary gear **56A** and the second transmission planetary gear **56B** are, for example, molded integrally. The first transmission planetary gear **56A** and the second transmission planetary gear **56B** can be formed separately and configured to rotate integrally with each other.

(71) The transmission planetary gears **48A** include, for example, the at least one second planetary gear **64**. The at least one second planetary gear **64** includes, for example, a plurality of second planetary gears **64**. The at least one second planetary gear **64** includes, for example, the second planetary gears **64** arranged at intervals about the rotational center axis **C2** of the shaft member **24**. The second planetary gears **64** are arranged at intervals about the rotational center axis **C2** of the shaft member **24**. The number of the second planetary gears **64** is, for example, two or greater and eight or less. In the present embodiment, the number of the second planetary gears **64** is four.

(72) The at least one second planetary gear **64** includes, for example, the second planetary gears **64** of different diameters. The at least one second planetary gear **64** includes a third transmission planetary gear **64A** and a fourth transmission planetary gear **64B** having a larger pitch circle diameter than the third transmission planetary gear **64A**. The third transmission planetary gear **64A** and the fourth transmission planetary gear **64B** are, for example, formed integrally. The third transmission planetary gear **64A** and the fourth transmission planetary gear **64B** can be formed separately and configured to rotate integrally with each other.

(73) In an example, at least the one of the first planetary gears **56** having the maximum diameter is arranged overlapping the at least one second planetary gear **64** as viewed in the axial direction **A1** of the shaft member **24**. The one of the first planetary gears **56** having the maximum diameter includes, for example, the second transmission planetary gear **56B**. Since at least the one of the first planetary gears **56** having the maximum diameter is arranged overlapping the at least one second

planetary gear **64** as viewed in the axial direction **A1**, the transmission device **22** is reduced in size in the radial direction with respect to the rotational center axis **C2**. Since at least one of the first planetary gears **56** having the maximum diameter overlaps the at least one second planetary gear **64** as viewed in the axial direction **A1**, the housing **30** is reduced in size in the radial direction of the input rotational shaft **28**.

(74) All of the first planetary gears **56** can be, for example, arranged overlapping the second planetary gears **64** as viewed in the axial direction **A1**. All of the first planetary gears **56** can be arranged so as not to overlap any of the second planetary gears **64** as viewed in the axial direction **A1**.

(75) In an example, at least the one of the second planetary gears **64** having the maximum diameter is arranged overlapping the at least one first planetary gear **56** as viewed in the axial direction **A1** of the shaft member **24**. The one of the second planetary gears **64** having the maximum diameter includes, for example, the fourth transmission planetary gear **64B**. Since at least the one of the second planetary gears **64** having the maximum diameter is arranged overlapping the at least one first planetary gear **56** as viewed in the axial direction **A1**, the transmission device **22** is reduced in size in the radial direction with respect to the rotational center axis **C2**. Since at least the one of the second planetary gears **64** having the maximum diameter is arranged overlapping the at least one first planetary gear **56** as viewed in the axial direction **A1**, the housing **30** is reduced in size in the radial direction of the input rotational shaft **28**.

(76) All of the second planetary gears **64** can be, for example, arranged overlapping the first planetary gears **56** as viewed in the axial direction **A1**. All of the second planetary gears **64** can be arranged so as not to overlap any of the first planetary gears **56** as viewed in the axial direction **A1**.

(77) The transmission carriers **48B** include, for example, the first carrier **58**. The first carrier **58** includes, for example, the first carrier portion **72** and a second carrier portion **74**. The input rotational portion **54** is, for example, connected to the first carrier **58**. The input rotational portion **54** is, for example, formed integrally with the first carrier portion **72**.

(78) The first carrier **58** includes, for example, at least one first carrier pin **58A** and supports the at least one first planetary gear **56** with the at least one first carrier pin **58A**. The at least one first planetary gear **56** is, for example, disposed between the first carrier portion **72** and the second carrier portion **74** in the axial direction **A1** of the shaft member **24**. The at least one first planetary gear **56** is, for example, rotatably supported by the at least one first carrier pin **58A**. The at least one first planetary gear **56** can be rotatably supported by the at least one first carrier pin **58A** via a bearing.

(79) The at least one first carrier pin **58A** is, for example, supported by the first carrier portion **72** and the second carrier portion **74**. The at least one first carrier pin **58A** includes, for example, a plurality of first carrier pins **58A**. Each of the first carrier pins **58A** includes two ends in the axial direction **A1** defining a first pin end **58B** and a second pin end **58C**. The first pin end **58B** is, for example, supported by the first carrier portion **72**. The second pin end **58C** is, for example, supported by the second carrier portion **74**.

(80) In an example, the first pin end **58B** can be fixed to the first carrier portion **72**, and the second pin end **58C** can be fixed to the second carrier portion **74**. In an example, the first pin end **58B** can be rotatably supported by the first carrier portion **72**, and the second pin end **58C** is rotatably supported by the second carrier portion **74**. In a case where the first pin end **58B** is rotatably supported by the first carrier portion **72** and the second pin end **58C** is rotatably supported by the second carrier portion **74**, the at least one first planetary gear **56** can be non-rotatably supported by the at least one first carrier pin **58A**.

(81) The transmission carriers **48B** include, for example, the second carrier **66**. The second carrier **66** includes, for example, the first carrier portion **72** and the second carrier portion **74**. A part of the first carrier **58** is, for example, formed integrally with a part of the second carrier **66**. A part of the first carrier **58** is, for example, molded integrally with a part of the second carrier **66**. The first

carrier portion **72** is, for example, a part of the first carrier **58** and a part of the second carrier **66**. The second carrier portion **74** is, for example, a part of the first carrier **58** and a part of the second carrier **66**.

(82) The second carrier **66** includes, for example, at least one second carrier pin **66A**, and supports the at least one second planetary gear **64** with the at least one second carrier pin **66A**. The at least one second planetary gear **64** is, for example, disposed between the first carrier portion **72** and the second carrier portion **74** in the axial direction **A1** of the shaft member **24**. The at least one second planetary gear **64** is, for example, rotatably supported by the at least one second carrier pin **66A**. The at least one second planetary gear **64** can be rotatably supported by the at least one second carrier pin **66A** via a bearing.

(83) The at least one second carrier pin **66A** is, for example, supported by the first carrier portion **72** and the second carrier portion **74**. The at least one second carrier pin **66A** includes, for example, a plurality of second carrier pins **66A**. The second carrier pins **66A** each include two ends in the axial direction **A1** defining a third pin end **66B** and a fourth pin end **66C**. The third pin end **66B** is, for example, supported by the first carrier portion **72**. The fourth pin end **66C** is, for example, supported by the second carrier portion **74**. In an example, the third pin end **66B** can be fixed to the first carrier portion **72**, and the fourth pin end **66C** can be fixed to the second carrier portion **74**. In an example, the third pin end **66B** can be rotatably supported by the first carrier portion **72**, and the fourth pin end **66C** can be rotatably supported by the second carrier portion **74**. In a case where the third pin end **66B** is rotatably supported by the first carrier portion **72** and the fourth pin end **66C** is rotatably supported by the second carrier portion **74**, the at least one second planetary gear **64** can be non-rotatably supported by the at least one second carrier pin **66A**.

(84) The second carrier **66** can include, for example, a third carrier portion **76** and at least one of the first carrier portion **72** and the second carrier portion **74**. In a case where the second carrier **66** includes the third carrier portion **76** and at least one of the first carrier portion **72** and the second carrier portion **74**, the at least one second carrier pin **66A** is supported by the third carrier portion **76** and at least one of the first carrier portion **72** and the second carrier portion **74**. The second carrier portion **74** is, for example, disposed between the first carrier portion **72** and the third carrier portion **76** in the axial direction **A1** of the shaft member **24**. An intermediate portion of the at least one second carrier pin **66A** is, for example, supported by the second carrier portion **74**.

(85) In a case where the second carrier **66** includes the third carrier portion **76** and at least one of the first carrier portion **72** and the second carrier portion **74**, for example, the third pin end **66B** is supported by the first carrier portion **72**, and the fourth pin end **66C** is supported by the third carrier portion **76**. The at least one first planetary gear **56** is, for example, disposed between the first carrier portion **72** and the second carrier portion **74**. The at least one second planetary gear **64** is, for example, disposed between the second carrier portion **74** and the third carrier portion **76**.

(86) The at least one first carrier pin **58A** is, for example, a member separate from the at least one second carrier pin **66A** and arranged at a distance from the at least one second carrier pin **66A**. The at least one first carrier pin **58A** includes, for example, the first carrier pins **58A** corresponding to the first planetary gears **56**. The at least one second carrier pin **66A** includes, for example, the second carrier pins **66A** corresponding to the second planetary gears **64**. The first carrier pins **58A** and the second carrier pins **66A** are, for example, arranged alternately at equal intervals about the rotational center axis **C2** of the shaft member **24**.

(87) In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one first carrier pin **58A** and the rotational center axis **C2** of the shaft member **24** is, for example, equal to the distance between the at least one second carrier pin **66A** and the rotational center axis **C2** of the shaft member **24**. In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one first carrier pin **58A** and the rotational center axis **C2** of the shaft member **24** can differ from the distance between the at least one second carrier pin **66A** and the rotational center axis **C2** of the

shaft member **24**.

(88) The at least one first carrier pin **58A** and the at least one second carrier pin **66A** are, for example, arranged overlapping a first plane that extends perpendicular to the shaft member **24**. The first plane, for example, extends perpendicular to the shaft member **24** and includes the at least one first carrier pin **58A** and the at least one second carrier pin **66A**. For example, the entirety of a plane that extends perpendicular to the shaft member **24** and includes the at least one first carrier pin **58A** and the at least one second carrier pin **66A** corresponds to the first plane. FIG. **15** is a cross-sectional view of the transmission device **22** taken along an example of the first plane.

(89) The transmission sun gears **48C** include, for example, the at least one first sun gear **60** and the at least one second sun gear **68**. The at least one first sun gear **60** is, for example, meshed with the at least one first planetary gear **56**. The at least one first sun gear **60** includes, for example, a first transmission sun gear **60A**. The first transmission sun gear **60A** is, for example, meshed with the second transmission planetary gear **56B**.

(90) The at least one second sun gear **68** is, for example, meshed with the at least one second planetary gear **64**. The at least one second sun gear **68** includes, for example, a second transmission sun gear **68A**. The second transmission sun gear **68A** is, for example, meshed with the third transmission planetary gear **64A**. The rotational center axis of the at least one first sun gear **60** and the rotational center axis of the at least one second sun gear **68** are, for example, substantially equal to the rotational center axis **C2** of the shaft member **24**.

(91) The at least one first sun gear **60** and the at least one second sun gear **68** are, for example, formed integrally. The first transmission sun gear **60A** and the second transmission sun gear **68A** are, for example, molded integrally. The at least one first sun gear **60** and the at least one second sun gear **68** can be formed separately and configured to rotate integrally.

(92) The at least one first sun gear **60** and the at least one second sun gear **68** are, for example, formed separately from the shaft member **24**, and configured to rotate integrally with the shaft member **24**. The first transmission sun gear **60A** and the second transmission sun gear **68A** are, for example, formed on an outer circumference of a first tubular member. For example, a first engagement portion **71A** is formed on an inner circumference of the first tubular member. The first engagement portion **71A** includes, for example, splines or serrations. The first engagement portion **71A** engages, for example, a first shaft member engagement **24A** of the shaft member **24**. The first shaft member engagement **24A** includes, for example, splines or serrations. The first shaft member engagement **24A** is, for example, molded integrally with the shaft member **24**. The at least one first sun gear **60** and the at least one second sun gear **68** can be molded integrally with the shaft member **24**.

(93) The at least one first sun gear **60** and the at least one second sun gear **68** are, for example, configured to rotate integrally in at least a first direction **B1** about the rotational center axis **C2** of the shaft member **24**. The at least one first sun gear **60** and the at least one second sun gear **68** can be configured to rotate integrally in a second direction **B2**, which is opposite to the first direction **B1**, about the rotational center axis **C2** of the shaft member **24**.

(94) The transmission ring gears **48D** include, for example, the at least one first ring gear **62**. The at least one first ring gear **62** is, for example, meshed with the at least one first planetary gear **56**. The at least one first ring gear **62** includes, for example, a plurality of first ring gears **62** that are rotatable independently from each other. The first planetary gears **56** are, for example, each meshed with first ring gears **62**.

(95) The first ring gears **62** include, for example, a first transmission ring gear **62A** and a second transmission ring gear **62B**. The first transmission ring gear **62A** is, for example, meshed with the first transmission planetary gear **56A**. The second transmission ring gear **62B** is, for example, meshed with the second transmission planetary gear **56B**.

(96) The transmission ring gears **48D** include, for example, the at least one second ring gear **70**. The at least one second ring gear **70** is, for example, meshed with the at least one second planetary

gear **64**. The at least one second ring gear **70** includes, for example, a plurality of second ring gears **70** that are rotatable independently from each other. The second planetary gears **64** are, for example, each meshed with second ring gears **70**.

(97) The second ring gears **70** include, for example, a third transmission ring gear **70A** and a fourth transmission ring gear **70B**. The third transmission ring gear **70A** is, for example, meshed with the third transmission planetary gear **64A**. The fourth transmission ring gear **70B** is, for example, meshed with the fourth transmission planetary gear **64B**.

(98) The transmission device **22** further includes, for example, a third planetary gear unit **77** provided on the shaft member **24**. The transmission device **22** includes, for example, a fourth planetary gear unit **82** provided on the shaft member **24** and a fifth planetary gear unit **87** provided on the shaft member **24**. The at least one secondary planetary gear group **48** further includes, for example, the third planetary gear unit **77**, the fourth planetary gear unit **82**, and the fifth planetary gear unit **87**.

(99) The third planetary gear unit **77** includes, for example, at least one third planetary gear **78**, a third carrier **79**, at least one third sun gear **80**, and at least one third ring gear **81**.

(100) The fourth planetary gear unit **82** includes, for example, at least one fourth planetary gear **83**, a fourth carrier **84**, at least one fourth sun gear **85**, and at least one fourth ring gear **86**.

(101) The fifth planetary gear unit **87** includes, for example, at least one fifth planetary gear **88**, a fifth carrier **89**, at least one fifth sun gear **90**, and at least one fifth ring gear **91**.

(102) The transmission planetary gears **48A** include, for example, the at least one third planetary gear **78**. The at least one third planetary gear **78** includes, for example, a plurality of third planetary gears **78** arranged at intervals about the rotational center axis C2 of the shaft member **24**. The at least one third planetary gears **78** are arranged at intervals about the rotational center axis C2 of the shaft member **24**. The number of the at least one third planetary gears **78** is, for example, two or greater and eight or less. In the present embodiment, the number of the at least one third planetary gears **78** is four.

(103) The at least one third planetary gears **78** include, for example, a fifth transmission planetary gear **78A** and a sixth transmission planetary gear **78B** having a smaller pitch circle diameter than the fifth transmission planetary gear **78A**. The fifth transmission planetary gear **78A** and the sixth transmission planetary gear **78B** are, for example, molded integrally. The fifth transmission planetary gear **78A** and the sixth transmission planetary gear **78B** are formed separately and configured to rotate integrally with each other.

(104) The transmission planetary gears **48A** include, for example, the at least one fourth planetary gear **83**. The at least one fourth planetary gear **83** includes, for example, a plurality of fourth planetary gears **83** arranged at intervals about the rotational center axis C2 of the shaft member **24**. The fourth planetary gears **83** are arranged at intervals about the rotational center axis C2 of the shaft member **24**. The number of the fourth planetary gears **83** is, for example, two or more and eight or less. In the present embodiment, the number of the fourth planetary gears **83** is four. The fourth planetary gears **83** include, for example, a seventh transmission planetary gear **83A**. The seventh transmission planetary gear **83A** has a pitch circle diameter that is, for example, smaller than the pitch circle diameter of the fifth transmission planetary gear **78A** and greater than the pitch circle diameter of the sixth transmission planetary gear **78B**.

(105) The transmission planetary gears **48A** include, for example, the at least one fifth planetary gear **88**. The at least one fifth planetary gear **88** includes, for example, a plurality of fifth planetary gears **88** arranged at intervals about the rotational center axis C2 of the shaft member **24**. The fifth planetary gears **88** are arranged at intervals about the rotational center axis C2 of the shaft member **24**. The number of the fifth planetary gears **88** is, for example, two or more and eight or less. In the present embodiment, the number of the fifth planetary gears **88** is four.

(106) The fifth planetary gears **88** include, for example, an eighth transmission planetary gear **88A** and a ninth transmission planetary gear **88B** having a larger pitch circle diameter than the eighth

transmission planetary gear **88A**. The eighth transmission planetary gear **88A** and the ninth transmission planetary gear **88B** are, for example, molded integrally. The eighth transmission planetary gear **88A** and the ninth transmission planetary gear **88B** are formed separately and configured to rotate integrally with each other.

(107) In an example, at least one of the at least one third planetary gear **78** having the maximum diameter is arranged overlapping at least one of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one fourth planetary gear **83**, and the at least one fifth planetary gear **88** as viewed in the axial direction **A1** of the shaft member **24**. The one of the at least one third planetary gears **78** having the maximum diameter includes, for example, the fifth transmission planetary gear **78A**. All of the at least one third planetary gear **78** can be arranged so as not to overlap any of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one fourth planetary gear **83**, and the at least one fifth planetary gear **88** as viewed in the axial direction **A1**.

(108) At least one of the at least one fourth planetary gear **83** having the maximum diameter is, for example, arranged overlapping at least one of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one third planetary gear **78**, and the at least one fifth planetary gear **88** as viewed in the axial direction **A1**. The one of the at least one fourth planetary gear **83** having the maximum diameter includes the seventh transmission planetary gear **83A**. All of the at least one fourth planetary gear **83** can be arranged so as not to overlap any of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one third planetary gear **78**, and the at least one fifth planetary gear **88** as viewed in the axial direction **A1**.

(109) At least one of the at least one fifth planetary gear **88** having the maximum diameter is, for example, arranged overlapping at least one of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one third planetary gear **78**, and the at least one fourth planetary gear **83** as viewed in the axial direction **A1**. The one of the at least one fifth planetary gear **88** having the maximum diameter includes, for example, the ninth transmission planetary gear **88B**. All of the at least one fifth planetary gear **88** can be arranged so as not to overlap any of the at least one first planetary gear **56**, the at least one second planetary gear **64**, the at least one third planetary gear **78**, and the at least one fourth planetary gear **83** as viewed in the axial direction **A1**.

(110) The transmission carriers **48B** further includes, for example, the third carrier **79**. The third carrier **79** includes, for example, the third carrier portion **76** and a fourth carrier portion **92**. The at least one third planetary gear **78** is, for example, disposed between the third carrier portion **76** and the fourth carrier portion **92** in the axial direction **A1**.

(111) The third carrier **79** includes, for example, at least one third carrier pin **79A** and supports the at least one third planetary gear **78** with the at least one third carrier pin **79A**. The at least one third carrier pin **79A** can be, for example, formed integrally with the at least one first carrier pin **58A** or the at least one second carrier pin **66A**. The at least one third planetary gear **78** is, for example, rotatably supported by the at least one third carrier pin **79A**. The at least one third planetary gear **78** can be rotatably supported by the at least one third carrier pin **79A** via a bearing.

(112) The at least one third carrier pin **79A** is, for example, supported by the third carrier portion **76** and the fourth carrier portion **92**. The at least one third carrier pin **79A** includes, for example, a plurality of third carrier pins **79A**. The third carrier pins **79A** each include two ends in the axial direction **A1** defining a fifth pin end **79B** and a sixth pin end **79C**. The fifth pin end **79B** is, for example, supported by the third carrier portion **76**. The sixth pin end **79C** is, for example, supported by the fourth carrier portion **92**.

(113) In an example, the fifth pin end **79B** can be fixed to the third carrier portion **76**, and the sixth pin end **79C** can be fixed to the fourth carrier portion **92**. In an example, the fifth pin end **79B** can be rotatably supported by the third carrier portion **76**, and the sixth pin end **79C** can be rotatably supported by the fourth carrier portion **92**. In a case where the fifth pin end **79B** is fixed to the third carrier portion **76** and the sixth pin end **79C** is rotatably supported by the fourth carrier portion **92**,

the at least one third planetary gear **78** can be non-rotatably supported by the at least one third carrier pin **79A**.

(114) The transmission carriers **48B** further include, for example, the fourth carrier **84**. The fourth carrier **84** includes, for example, the third carrier portion **76** and the fourth carrier portion **92**. The at least one fourth planetary gear **83** is, for example, disposed between the third carrier portion **76** and the fourth carrier portion **92** in the axial direction **A1**.

(115) The fourth carrier **84** includes, for example, at least one fourth carrier pin **84A** and supports the at least one fourth planetary gear **83** with the at least one fourth carrier pin **84A**. The at least one fourth planetary gear **83** is, for example, rotatably supported by the at least one fourth carrier pin **84A**. The at least one fourth planetary gear **83** can be rotatably supported by the at least one fourth carrier pin **84A** via a bearing.

(116) The at least one fourth carrier pin **84A** is, for example, supported by the third carrier portion **76** and the fourth carrier portion **92**. The at least one fourth carrier pin **84A** includes, for example, a plurality of fourth carrier pins **84A**. The fourth carrier pins **84A** each include two ends in the axial direction **A1** defining a seventh pin end **84B** and an eighth pin end **84C**. The seventh pin end **84B** is, for example, supported by the third carrier portion **76**. The eighth pin end **84C** is, for example, supported by the fourth carrier portion **92**.

(117) In an example, the seventh pin end **84B** can be fixed to the third carrier portion **76**, and the eighth pin end **84C** can be fixed to the fourth carrier portion **92**. In an example, the seventh pin end **84B** can be rotatably supported by the third carrier portion **76**, and the eighth pin end **84C** can be rotatably supported by the fourth carrier portion **92**. In a case where the seventh pin end **84B** is rotatably supported by the third carrier portion **76** and the eighth pin end **84C** is rotatably supported by the fourth carrier portion **92**, the at least one fourth planetary gear **83** can be non-rotatably supported by the at least one fourth carrier pin **84A**.

(118) The transmission carriers **48B** further include, for example, the fifth carrier **89**. The fifth carrier **89** includes, for example, the third carrier portion **76**, the fourth carrier portion **92**, and a fifth carrier portion **93**. The output rotational portion **32** includes, for example, the fifth carrier **89**. The output rotational portion **32** is, for example, molded integrally with the fifth carrier portion **93**. The output rotational portion **32** and the fifth carrier portion **93** can be formed separately and configured to rotate integrally with each other.

(119) The fifth carrier **89** includes, for example, at least one fifth carrier pin **89A** and supports the at least one fifth planetary gear **88** with the at least one fifth carrier pin **89A**. The at least one fifth planetary gear **88** is, for example, disposed between the fourth carrier portion **92** and the fifth carrier portion **93** in the axial direction **A1**. The at least one fifth planetary gear **88** is, for example, rotatably supported by the at least one fifth carrier pin **89A**. The at least one fifth planetary gear **88** can be rotatably supported by the at least one fifth carrier pin **89A** via a bearing.

(120) In an example, the third carrier **79** can be the same as the fourth carrier **84**. The third carrier **79** can differ from the fourth carrier **84**. Part of the third carrier **79** is, for example, formed integrally with part of the fifth carrier **89**. Part of the fourth carrier **84** is formed integrally with part of the fifth carrier **89**. Part of the fourth carrier **84** can be molded integrally with part of the fifth carrier **89**.

(121) The at least one fifth carrier pin **89A** is, for example, supported by the third carrier portion **76**, the fourth carrier portion **92**, and the fifth carrier portion **93**. The at least one fifth carrier pin **89A** includes, for example, a plurality of fifth carrier pins **89A**. The fifth carrier pins **89A** each include two ends in the axial direction **A1** defining a ninth pin end **89B** and a tenth pin end **89C**. The ninth pin end **89B** is, for example, supported by the third carrier portion **76**. The tenth pin end **89C** is, for example, supported by the fifth carrier portion **93**. The fourth carrier portion **92** is, for example, disposed between the third carrier portion **76** and the fifth carrier portion **93**. The fourth carrier portion **92** supports, for example, a portion of the at least one fifth carrier pin **89A** between the third carrier portion **76** and the fifth carrier portion **93**.

(122) In an example, the ninth pin end **89B** can be fixed to the third carrier portion **76**, and the tenth pin end **89C** can be fixed to the fifth carrier portion **93**. In an example, the ninth pin end **89B** can be rotatably supported by the third carrier portion **76**, and the tenth pin end **89C** can be rotatably supported by the fifth carrier portion **93**. In a case where the ninth pin end **89B** is rotatably supported by the third carrier portion **76** and the tenth pin end **89C** is rotatably supported by the fifth carrier portion **93**, the at least one fifth planetary gear **88** can be non-rotatably supported by the at least one fifth carrier pin **89A**.

(123) The at least one third carrier pin **79A** is, for example, a member molded separately from the at least one fifth carrier pin **89A** and arranged at a distance from the at least one fifth carrier pin **89A**. The at least one third carrier pin **79A** includes, for example, the third carrier pins **79A** corresponding to the at least one third planetary gears **78**. The at least one fifth carrier pin **89A** includes, for example, the fifth carrier pins **89A** corresponding to the fifth planetary gears **88**. The third carrier pins **79A** and the fifth carrier pins **89A** are, for example, arranged at equal intervals about the rotational center axis **C2** of the shaft member **24**.

(124) The at least one fourth carrier pin **84A** is, for example, a member molded separately from the at least one fifth carrier pin **89A** and arranged at a distance from the at least one fifth carrier pin **89A**. The at least one fourth carrier pin **84A** includes, for example, a plurality of the fourth carrier pins **84A** corresponding to the fourth planetary gears **83**. The fourth carrier pins **84A** and the fifth carrier pins **89A** are, for example, arranged at equal intervals about the rotational center axis **C2** of the shaft member **24**.

(125) In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one third carrier pin **79A** and the rotational center axis **C2** of the shaft member **24** is, for example, substantially equal to the distance between the at least one fifth carrier pin **89A** and the rotational center axis **C2** of the shaft member **24**. In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one third carrier pin **79A** and the rotational center axis **C2** of the shaft member **24** can differ from the distance between the at least one fifth carrier pin **89A** and the rotational center axis **C2** of the shaft member **24**.

(126) In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one fourth carrier pin **84A** and the rotational center axis **C2** of the shaft member **24** is, for example, substantially equal to the distance between the at least one fifth carrier pin **89A** and the rotational center axis **C2** of the shaft member **24**. In a plane that extends perpendicular to the rotational center axis **C2** of the shaft member **24**, the distance between the at least one fourth carrier pin **84A** and the rotational center axis **C2** of the shaft member **24** can differ from the distance between the at least one fifth carrier pin **89A** and the rotational center axis **C2** of the shaft member **24**.

(127) The third carrier pin **79A** and the at least one fifth carrier pin **89A** are, for example, arranged overlapping a second plane that extends perpendicular to the shaft member **24**. The at least one fourth carrier pin **84A** and the at least one fifth carrier pin **89A** are, for example, arranged overlapping the second plane that extends perpendicular to the shaft member **24**. The second plane, for example, extends perpendicular to the shaft member **24** and includes the at least one fourth carrier pin **84A** and the at least one fifth carrier pin **89A**. For example, the entirety of a plane that extends perpendicular to the shaft member **24** and includes the at least one fourth carrier pin **84A** and the at least one fifth carrier pin **89A** corresponds to the second plane. FIG. **16** shows a cross-sectional view of the transmission device **22** in an example of the second plane.

(128) The transmission sun gears **48C** include, for example, at least one third sun gear **80**, the at least one fourth sun gear **85**, and the at least one fifth sun gear **90**. The rotational center axis of the at least one third sun gear **80**, the rotational center axis of the at least one fourth sun gear **85**, and the rotational center axis of the at least one fifth sun gear **90** are, for example, substantially equal to the rotational center axis **C2** of the shaft member **24**.

(129) The at least one third sun gear **80** is, for example, meshed with the at least one third planetary gear **78**. The at least one third sun gear **80** includes, for example, a third transmission sun gear **80A**. The third transmission sun gear **80A** is, for example, meshed with the sixth transmission planetary gear **78B**.

(130) The at least one fourth sun gear **85** is, for example, meshed with the at least one fourth planetary gear **83**. The at least one fourth sun gear **85** includes, for example, a fourth transmission sun gear **85A**. The fourth transmission sun gear **85A** is, for example, meshed with the seventh transmission planetary gear **83A**.

(131) The at least one fifth sun gear **90** is, for example, meshed with the at least one fifth planetary gear **88**. The at least one fifth sun gear **90** includes, for example, a fifth transmission sun gear **90A**. The at least one fifth sun gear **90** is, for example, meshed with the ninth transmission planetary gear **88B**.

(132) The at least one third sun gear **80** and the at least one fourth sun gear **85** are, for example, molded integrally. The third transmission sun gear **80A** and the fourth transmission sun gear **85A** are, for example, molded integrally. The at least one first sun gear **60** and the at least one second sun gear **68** can be formed separately and configured to rotate integrally. The third transmission sun gear **80A** and the fourth transmission sun gear **85A** can be formed separately and configured to rotate integrally with each other.

(133) The at least one third sun gear **80** and the at least one fourth sun gear **85** are, for example, formed separately from the shaft member **24** and configured to rotate integrally with the shaft member **24**. The third transmission sun gear **80A** and the fourth transmission sun gear **85A** are, for example, formed on an outer circumference of a second tubular member. For example, a second engagement portion **71B** is formed on an inner circumference of the second tubular member. The second engagement portion **71B** includes, for example, splines or serrations. The second engagement portion **71B** engages, for example, a second shaft member engagement **24B** of the shaft member **24**. The second shaft member engagement **24B** includes, for example, splines or serrations. The second shaft member engagement **24B** is, for example, molded integrally with the shaft member **24**. The at least one third sun gear **80** and the at least one fourth sun gear **85** can be, for example, molded integrally with the shaft member **24**.

(134) The at least one fifth sun gear **90** is, for example, molded integrally with the shaft member **24**. The at least one fifth sun gear **90** can be formed separately from the shaft member **24**, and configured to rotate integrally with the shaft member **24** using splines or serrations in the same manner as the at least one third sun gear **80** and the at least one fourth sun gear **85**.

(135) The at least one third sun gear **80** and the at least one fifth sun gear **90** are, for example, configured to rotate integrally in at least the first direction **B1** about the rotational center axis **C2** of the shaft member **24**. The at least one fourth sun gear **85** and the at least one fifth sun gear **90** are, for example, configured to rotate integrally in at least the first direction **B1** about the rotational center axis **C2** of the shaft member **24**. The at least one first sun gear **60**, the at least one second sun gear **68**, the at least one fourth sun gear **85**, and the at least one fifth sun gear **90** are, for example, configured to rotate integrally in at least the first direction **B1** about the rotational center axis **C2** of the shaft member **24**.

(136) The at least one first sun gear **60**, the at least one second sun gear **68**, the at least one third sun gear **80**, the at least one fourth sun gear **85**, and the at least one fifth sun gear **90** are, for example, configured to rotate integrally in at least the first direction **B1** about the rotational center axis **C2** of the shaft member **24**. In addition, the at least one first sun gear **60**, the at least one second sun gear **68**, the at least one third sun gear **80**, the at least one fourth sun gear **85**, and the at least one fifth sun gear **90** can be configured to rotate integrally in the second direction **B2**, opposite to the first direction **B1**, about the rotational center axis **C2** of the shaft member **24**.

(137) The first transmission sun gear **60A** has, for example, a larger pitch circle diameter than the fifth transmission sun gear **90A**. The first transmission sun gear **60A** has, for example, a smaller

pitch circle diameter than each of the second transmission sun gear **68A**, the third transmission sun gear **80A**, and the fourth transmission sun gear **85A**. The second transmission sun gear **68A** has, for example, a larger pitch circle diameter than each of the first transmission sun gear **60A** and the fifth transmission sun gear **90A**. The second transmission sun gear **68A** has, for example, a smaller pitch circle diameter than each of the third transmission sun gear **80A** and the fourth transmission sun gear **85A**.

(138) The third transmission sun gear **80A** has, for example, a larger pitch circle diameter than each of the first transmission sun gear **60A**, the second transmission sun gear **68A**, the fourth transmission sun gear **85A**, and the fifth transmission sun gear **90A**. The fourth transmission sun gear **85A** has, for example, a larger pitch circle diameter than each of the first transmission sun gear **60A**, the second transmission sun gear **68A**, and the fifth transmission sun gear **90A**. The fourth transmission sun gear **85A** has, for example, a smaller pitch circle diameter than the third transmission sun gear **80A**. The fifth transmission sun gear **90A** has, for example, a smaller pitch circle diameter than each of the first transmission sun gear **60A**, the second transmission sun gear **68A**, the third transmission sun gear **80A**, and the fourth transmission sun gear **85A**.

(139) The transmission ring gears **48D** include, for example, at least one third ring gear **81**. The at least one third ring gear **81** is, for example, meshed with the at least one third planetary gear **78**. The at least one third ring gear **81** includes, for example, a fifth transmission ring gear **81A**. The fifth transmission ring gear **81A** is, for example, meshed with the fifth transmission planetary gear **78A**.

(140) The transmission ring gears **48D** include, for example, the at least one fourth ring gear **86**. The at least one fourth ring gear **86** is, for example, meshed with the at least one fourth planetary gear **83**. The at least one fourth ring gear **86** includes, for example, a sixth transmission ring gear **86A**. The sixth transmission ring gear **86A** is, for example, meshed with the seventh transmission planetary gear **83A**.

(141) The transmission ring gears **48D** include, for example, the at least one fifth ring gear **91**. The at least one fifth ring gear **91** is, for example, meshed with the at least one fifth planetary gear **88**. The at least one fifth ring gear **91** includes, for example, a seventh transmission ring gear **91A**. The seventh transmission ring gear **91A** is, for example, meshed with the eighth transmission planetary gear **88A**.

(142) In the transmission device **22**, for example, a rotational state of at least one of the at least one first sun gear **60**, the at least one second sun gear **68**, the at least one first ring gear **62**, and the at least one second ring gear **70** is controlled to switch the transmission stage in steps. In the transmission device **22**, for example, a rotational state of at least one of the at least one third sun gear **80** and the at least one third ring gear **81** is controlled to switch the transmission stage in steps. In the transmission device **22**, for example, a rotational state of at least one of the at least one fourth sun gear **85**, the at least one fifth sun gear **90**, the at least one fourth ring gear **86**, and the at least one fifth ring gear **91** is controlled to switch the transmission stage in steps.

(143) In the transmission device **22**, for example, a rotational state of at least one of the at least one first ring gear **62**, the at least one second ring gear **70**, the at least one third ring gear **81**, the at least one fourth ring gear **86**, and the at least one fifth ring gear **91** is controlled to switch the transmission stage in steps. In the transmission device **22**, a rotational state of at least one of the at least one first sun gear **60**, the at least one second sun gear **68**, the at least one third sun gear **80**, the at least one fourth sun gear **85**, and the at least one fifth sun gear **90** is controlled to switch the transmission stage in steps.

(144) The transmission device **22** includes, for example, a plurality of transmission paths. The transmission paths correspond to the transmission stages, respectively. For example, the transmission stage is switched by changing one transmission path to another transmission path. The transmission paths form part of the transmission path of the human driving force. The driving force that is input to the input rotational portion **54** from the speed-increasing output portion **46** is

transmitted through one of the transmission paths, and is output from the output rotational portion 32. The transmission paths include a first transmission path, a second transmission path, a third transmission path, a fourth transmission path, a fifth transmission path, a sixth transmission path, a seventh transmission path, an eighth transmission path, a ninth transmission path, a tenth transmission path, an eleventh transmission path, and a twelfth transmission path.

(145) In the first transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the second carrier 66, the fourth transmission planetary gear 64B, the second transmission sun gear 68A, the shaft member 24, the fifth transmission sun gear 90A, the ninth transmission planetary gear 88B, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(146) In the second transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the second carrier 66, the third transmission planetary gear 64A, the second transmission sun gear 68A, the shaft member 24, the fifth transmission sun gear 90A, the ninth transmission planetary gear 88B, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(147) In the third transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the first carrier 58, the second transmission planetary gear 56B, the first transmission sun gear 60A, the shaft member 24, the fifth transmission sun gear 90A, the ninth transmission planetary gear 88B, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(148) In the fourth transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the first carrier 58, the first transmission planetary gear 56A, the second transmission planetary gear 56B, the first transmission sun gear 60A, the shaft member 24, the fifth transmission sun gear 90A, the ninth transmission planetary gear 88B, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(149) In the fifth transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the second carrier 66, the fourth transmission planetary gear 64B, the second transmission sun gear 68A, the shaft member 24, the fourth transmission sun gear 85A, the seventh transmission planetary gear 83A, the fourth carrier 84, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(150) In the sixth transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the second carrier 66, the third transmission planetary gear 64A, the second transmission sun gear 68A, the shaft member 24, the fourth transmission sun gear 85A, the seventh transmission planetary gear 83A, the fourth carrier 84, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(151) In the seventh transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the first carrier 58, the second transmission planetary gear 56B, the first transmission sun gear 60A, the shaft member 24, the fourth transmission sun gear 85A, the seventh transmission planetary gear 83A, the fourth carrier 84, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(152) In the eighth transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the first carrier 58, the first transmission planetary gear 56A, the second transmission planetary gear 56B, the first transmission sun gear 60A, the shaft member 24, the fourth transmission sun gear 85A, the seventh transmission planetary gear 83A, the fourth carrier 84, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(153) In the ninth transmission path, for example, the driving force that is input to the input rotational portion 54 is transmitted through the second carrier 66, the fourth transmission planetary gear 64B, the second transmission sun gear 68A, the shaft member 24, the third transmission sun gear 80A, the sixth transmission planetary gear 78B, the fourth carrier 84, and the fifth carrier portion 93, and is output to the output rotational portion 32.

(163) For example, in a case where the second to sixth transmission ring gears **62B**, **70A**, **70B**, **81A**, and **86A** are in the permission state, and the first transmission ring gear **62A** and the seventh transmission ring gear **91A** are in the restriction state, the fourth transmission path is selected from the transmission paths.

(164) For example, in a case where the first to third transmission ring gears **62A**, **62B**, and **70A**, the fifth transmission ring gear **81A**, and the seventh transmission ring gear **91A** are in the permission state, and the fourth transmission ring gear **70B** and the sixth transmission ring gear **86A** are in the restriction state, the fifth transmission path is selected from the transmission paths.

(165) For example, in a case where the first transmission ring gear **62A**, the second transmission ring gear **62B**, the fourth transmission ring gear **70B**, the fifth transmission ring gear **81A**, and the seventh transmission ring gear **91A** are in the permission state, and the third transmission ring gear **70A** and the sixth transmission ring gear **86A** are in the restriction state, the sixth transmission path is selected from the transmission paths.

(166) For example, in a case where the first transmission ring gear **62A**, the third to fifth transmission ring gears **70A**, **70B**, and **81A**, and the seventh transmission ring gear **91A** are in the permission state, and the second transmission ring gear **62B** and the sixth transmission ring gear **86A** are in the restriction state, the seventh transmission path is selected from the transmission paths.

(167) For example, in a case where the second to fifth transmission ring gears **62B**, **70A**, **70B**, and **81A** and the seventh transmission ring gear **91A** are in the permission state, and the first transmission ring gear **62A** and the sixth transmission ring gear **86A** are in the restriction state, the eighth transmission path is selected from the transmission paths.

(168) For example, in a case where the first to third transmission ring gears **62A**, **62B**, and **70A**, the sixth transmission ring gear **86A**, and the seventh transmission ring gear **91A** are in the permission state, and the fourth transmission ring gear **70B** and the fifth transmission ring gear **81A** are in the restriction state, the ninth transmission path is selected from the transmission paths.

(169) For example, in a case where the first transmission ring gear **62A**, the second transmission ring gear **62B**, the fourth transmission ring gear **70B**, the sixth transmission ring gear **86A**, and the seventh transmission ring gear **91A** are in the permission state, and the third transmission ring gear **70A** and the fifth transmission ring gear **81A** are in the restriction state, the tenth transmission path is selected from the transmission paths.

(170) For example, in a case where the first transmission ring gear **62A**, the third transmission ring gear **70A**, the fourth transmission ring gear **70B**, the sixth transmission ring gear **86A**, and the seventh transmission ring gear **91A** are in the permission state, and the second transmission ring gear **62B** and the fifth transmission ring gear **81A** are in the restriction state, the eleventh transmission path is selected from the transmission paths.

(171) For example, the second to fourth transmission ring gears **62B**, **70A**, and **70B**, the sixth transmission ring gear **86A**, and the seventh transmission ring gear **91A** are in the permission state, and the first transmission ring gear **62A** and the fifth transmission ring gear **81A** are in the restriction state, the twelfth transmission path is selected from the transmission paths.

(172) The transmission device **22** further includes, for example, a transmission switching mechanism **94**. The transmission device **22** is, for example, configured to change the rotational state of the transmission ring gears **48D** to switch the transmission stages of the transmission device **22** in steps.

(173) The transmission switching mechanism **94**, for example, switches the rotational state of the first ring gears **62** to switch the transmission stage in steps. The transmission switching mechanism **94**, for example, switches the rotational state of the second ring gears **70** to switch the transmission stage in steps. The transmission switching mechanism **94**, for example, switches the rotational state of the at least one first ring gear **62** and the rotational state of the at least one second ring gear **70** to switch the transmission stage in steps.

(174) The transmission switching mechanism **94**, for example, is configured to switch the rotational state of the at least one third ring gear **81** to switch the transmission stage in steps. The transmission switching mechanism **94**, for example, is configured to switch the rotational state of the at least one fourth ring gear **86** to switch the transmission stage in steps. The transmission switching mechanism **94**, for example, is configured to switch the rotational state of the at least one fifth ring gear **91** to switch the transmission stage in steps. The transmission switching mechanism **94**, for example, is configured to switch the rotational states of the at least one first ring gear **62**, the at least one second ring gear **70**, the at least one third ring gear **81**, the at least one fourth ring gear **86**, and the at least one fifth ring gear **91** to switch the transmission stage in steps.

(175) The transmission switching mechanism **94** includes, for example, a plurality of clutches **95**. The clutches **95**, for example, switch the rotational states of the transmission ring gears **48D** between the permission state and the restriction state. Each of the clutches **95** includes, for example, a rotational member **96** and an engaging member **97** configured to engage at least one recess **78E** provided on an outer circumference of the transmission ring gears **48D**. The clutch **95** is, for example, configured to switch, as the rotational member **96** rotates, between a first state in which the engaging member **97** is separated from the at least one recess **78E** and a second state in which the engaging member **97** engages the at least one recess **78E**. The engaging member **97** includes, for example, a pawl. Preferably, the at least one recess **78E** includes a plurality of recesses **78E**. The recesses **78E** are, for example, arranged on the outer circumference of the transmission ring gears **48D** at intervals about the rotational center axis C2 of the shaft member **24**.

(176) The clutches **95** include, for example, a first clutch **95A** corresponding to the first transmission ring gear **62A**. The first clutch **95A** includes, for example, a first rotational member **96A** and a first engaging member **97A**. The first engaging member **97A** is, for example, configured to engage at least one recess **78E** of the first transmission ring gear **62A**. The first clutch **95A** is, for example, configured to switch, as the first rotational member **96A** rotates, between a first state in which the first engaging member **97A** is separated from the at least one recess **78E** and a second state in which the first engaging member **97A** engages the at least one recess **78E**.

(177) The clutches **95** include, for example, a second clutch **95B** corresponding to the second transmission ring gear **62B**. The second clutch **95B** includes, for example, a second rotational member **96B** and a second engaging member **97B**. The second engaging member **97B** is, for example, configured to engage the at least one recess **78E** of the second transmission ring gear **62B**. The second clutch **95B** is, for example, configured to switch, as the second rotational member **96B** rotates, between a first state in which the second engaging member **97B** is separated from the at least one recess **78E** and a second state in which the second engaging member **97B** engages the at least one recess **78E**.

(178) The clutches **95** include, for example, a third clutch **95C** corresponding to the third transmission ring gear **70A**. The third clutch **95C** includes, for example, a third rotational member **96C** and a third engaging member **97C**. The third engaging member **97C** is, for example, configured to engage at least one recess **78E** of the third transmission ring gear **70A**. The third clutch **95C** is, for example, configured to switch, as the third rotational member **96C** rotates, between a first state in which the third engaging member **97C** is separated from the at least one recess **78E** and a second state in which the third engaging member **97C** engages the at least one recess **78E**.

(179) The clutches **95** include, for example, a fourth clutch **95D** corresponding to the fourth transmission ring gear **70B**. The fourth clutch **95D** includes, for example, a fourth rotational member **96D** and a fourth engaging member **97D**. The fourth engaging member **97D** is, for example, configured to engage at least one recess **78E** of the fourth transmission ring gear **70B**. The fourth clutch **95D** is, for example, configured to switch, as the fourth rotational member **96D** rotates, between a first state in which the fourth engaging member **97D** is separated from the at least one recess **78E** and a second state in which the fourth engaging member **97D** engages the at

least one recess **78E**.

(180) The clutches **95** include, for example, a fifth clutch **95E** corresponding to the fifth transmission ring gear **81A**. The fifth clutch **95E** includes, for example, a fifth rotational member **96E** and a fifth engaging member **97E**. The fifth engaging member **97E** is, for example, configured to engage at least one recess **78E** of the fifth transmission ring gear **81A**. The fifth clutch **95E** is, for example, configured to switch, as the fifth rotational member **96E** rotates, between a first state in which the fifth engaging member **97E** is separated from the at least one recess **78E** and a second state in which the fifth engaging member **97E** engages the at least one recess **78E**.

(181) The clutches **95** include, for example, a sixth clutch **95F** corresponding to the sixth transmission ring gear **86A**. The sixth clutch **95F** includes, for example, a sixth rotational member **96F** and a sixth engaging member **97F**. The sixth engaging member **97F** is, for example, configured to engage at least one recess **78E** of the sixth transmission ring gear **86A**. The sixth clutch **95F** is, for example, configured to switch, as the sixth rotational member **96F** rotates, between a first state in which the sixth engaging member **97F** is separated from the at least one recess **78E** and a second state in which the sixth engaging member **97F** engages the at least one recess **78E**.

(182) The clutches **95** include, for example, a seventh clutch **95G** corresponding to the seventh transmission ring gear **91A**. The seventh clutch **95G** includes, for example, a seventh rotational member **96G** and a seventh engaging member **97G**. The seventh engaging member **97G** is, for example, configured to engage at least one recess **78E** of the seventh transmission ring gear **91A**. The seventh clutch **95G** is, for example, configured to switch, as the seventh rotational member **96G** rotates, between a first state in which the seventh engaging member **97G** is separated from the at least one recess **78E** and a second state in which the seventh engaging member **97G** engages the at least one recess **78E**.

(183) The transmission switching mechanism **94** includes, for example, an electric actuator. The electric actuator includes, for example, a transmission motor **94A**. The electric actuator is, for example, configured to switch the states of the clutches **95** between the first state and the second state. The rotational member **96** is, for example, configured to be rotated by the electric actuator.

(184) The transmission device **22** includes, for example, a first controller **22A**. The first controller **22A** is, for example, configured to control the transmission motor **94A**. Thus, the first controller **22A** can also be referred to as a transmission controller. In an example, the first controller **22A** includes a processor that executes a predetermined control program. The processor includes, for example, a central processing unit (CPU) or a micro processing unit (MPU). Thus, the first controller **22A** is formed of one or more semiconductor chips that are mounted on a circuit board. The processor can be provided at separate positions. The first controller **22A** can include one or more microcomputers. The term “controller” as used herein refers to hardware that executes a software program, and does not include a human being. Accordingly, the first controller **22A** can also be referred to as a first electronic controller, an electronic transmission controller, or merely as an electronic controller.

(185) The transmission device **22** includes, for example, first storage. The first storage stores various control programs and information used for various control processes. The first storage includes, for example, nonvolatile memory and volatile memory. A non-volatile memory includes, for example, at least one of a read-only memory (ROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), and a flash memory. The volatile memory includes, for example, a random-access memory (RAM). Thus, the first storage can be any computer storage device or any non-transitory computer-readable medium with the sole exception of a transitory, propagating signal.

(186) The first controller **22A** is, for example, configured to receive a transmission signal from a transmission operating device and control the transmission motor **94A**. The first controller **22A** can be connected to the transmission operating device by a wireless communication device or an electrical cable. Instead of receiving a transmission signal from the transmission operating device,

the first controller **22A** can receive a signal from a sensor mounted on the human-powered vehicle and control the transmission motor **94A** so that an automatic transmission is performed.

(187) For example, in a case where the first transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first to third clutches **95A**, **95B**, and **95C**, the fifth clutch **95E**, and the sixth clutch **95F** are in the first state. For example, in a case where the first transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the fourth clutch **95D** and the seventh clutch **95G** are in the second state.

(188) For example, in a case where the second transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A**, the second clutch **95B**, and the fourth to sixth clutches **95D**, **95E**, and **95F** are in the first state. For example, in a case where the second transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the third clutch **95C** and the seventh clutch **95G** are in the second state.

(189) For example, in a case where the third transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A** and the third to sixth clutches **95C**, **95D**, **95E**, and **95F** are in the first state. For example, in a case where the third transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second clutch **95B** and the seventh clutch **95G** are in the second state.

(190) For example, in a case where the fourth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second to sixth clutches **95B**, **95C**, **95D**, **95E**, and **95F** are in the first state. For example, in a case where the fourth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A** and the seventh clutch **95G** are in the second state.

(191) For example, in a case where the fifth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first to third clutches **95A**, **95B**, and **95C**, the fifth clutch **95E**, and the seventh clutch **95G** are in the first state. For example, in a case where the fifth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the fourth clutch **95D** and the sixth clutch **95F** are in the second state.

(192) For example, in a case where the sixth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A**, the second clutch **95B**, the fourth clutch **95D**, the fifth clutch **95E**, and the seventh clutch **95G** are in the first state. For example, in a case where the sixth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the third clutch **95C** and the sixth clutch **95F** are in the second state.

(193) For example, in a case where the seventh transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A**, the third clutch **95C**, the fourth clutch **95D**, the fifth clutch **95E**, and the seventh clutch **95G** are in the first state. For example, in a case where the seventh transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second clutch **95B** and the sixth clutch **95F** are in the second state.

(194) For example, in a case where the eighth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second to fifth clutches **95B**, **95C**, **95D**, and **95E** and the seventh clutch **95G** are in the first state. For example, in a case where the eighth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A** and the sixth clutch **95F** are in the second state.

(195) For example, in a case where the ninth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first to third clutches **95A**, **95B**, and **95C**, the sixth clutch **95F**, and the seventh clutch **95G** are in the first state. For example, in a case where the ninth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the fourth clutch **95D** and the fifth clutch **95E** are in the second state.

(196) For example, in a case where the tenth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A**, the second clutch **95B**, the fourth clutch **95D**, the sixth clutch **95F**, and the seventh clutch **95G** are in the first state. For example, in a case where the tenth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the third clutch **95C** and the fifth clutch **95E** are in the second state.

(197) For example, in a case where the eleventh transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A**, the third clutch **95C**, the fourth clutch **95D**, the sixth clutch **95F**, and the seventh clutch **95G** are in the first state. For example, in a case where the eleventh transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second clutch **95B** and the fifth clutch **95E** are in the second state.

(198) For example, in a case where the twelfth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the second to fourth clutches **95B**, **95C**, and **95D**, the sixth clutch **95F**, and the seventh clutch **95G** are in the first state. For example, in a case where the twelfth transmission path is selected from the transmission paths, the transmission controller **22A** controls the transmission motor **94A** so that the first clutch **95A** and the fifth clutch **95E** are in the second state.

Second Embodiment

(199) A second embodiment of the component **20** and the transmission device **22** will now be described with reference to FIGS. **19** to **21**. In the second embodiment, same reference characters are given to those elements of the component **20** and the transmission device **22** that are the same as the corresponding elements of the component **20** and the transmission device **22** of the first embodiment. Such elements will not be described in detail.

(200) In the second embodiment, the component **20** for a human-powered vehicle includes, for example, the transmission device **22** and a motor **98** configured to apply a propulsion force to the human-powered vehicle. In an example, the component **20** further includes the motor **98** configured to apply a propulsion force to the human-powered vehicle in addition to the transmission device **22**, the shaft member **24**, and the speed increaser **26**. The motor **98** is, for example, an assist motor. The motor **98** includes, for example, a motor rotational axle **98A**.

(201) The component **20** further includes, for example, a resultant force portion **99**. The resultant force portion **99**, for example, combines the motor driving force of the motor **98** with the human driving force. The resultant force portion **99** includes, for example, at least one of the speed-increasing input portion **44** and the input rotational portion **54**. The speed-increasing input portion **44** is, for example, configured to receive the rotational torque from the motor **98**. The speed-increasing input portion **44**, for example, combines the motor driving force of the motor **98** with the human driving force.

(202) The component **20** further includes, for example, a motor speed reducer **98B**. For example, a motor driving force of the motor **98** is input to the motor speed reducer **98B**. The motor speed reducer **98B**, for example, outputs the motor driving force of the motor **98** to the resultant force portion **99**. The motor speed reducer **98B**, for example, connects the motor rotational axle **98A** and the resultant force portion **99**. In a case where the rotational torque of the motor **98** is input to the speed-increasing input portion **44**, the reduction ratio of the motor speed reducer **98B** is decreased. This allows for reduction in size of the motor speed reducer **98B**.

(203) The motor speed reducer **98B** includes, for example, at least one of a gear, a pulley, and an endless annular member. The endless annular member includes, for example, at least one of a chain and a belt. The motor speed reducer **98B** includes, for example, a gear. The motor speed reducer **98B** can include a planetary gear mechanism. The motor speed reducer **98B** can include a pulley and a belt. The motor speed reducer **98B** can include a gear and a chain.

(204) The component **20** is, for example, a drive unit. The transmission device **22**, the shaft member **24**, the speed increaser **26**, at least part of the input rotational shaft **28**, at least part of the motor **98**, and at least part of the motor speed reducer **98B** are, for example, disposed in the inner cavity **20A** of the housing **30**.

(205) The component **20** further includes, for example, a third one-way clutch. The third one-way clutch is, for example, provided in a transmission path of the motor driving force between the motor **98** and the resultant force portion **99**. The third one-way clutch, for example, allows transmission of the rotational force of the motor **98** to the resultant force portion **99** and restricts transmission of the rotational force of the resultant force portion **99** caused by the human driving force to the motor **98**. The third one-way clutch includes, for example, at least one of a roller clutch, a pawl-type clutch, and a sprag-type clutch.

(206) The drive unit includes, for example, a second controller **100** configured to control the motor **98**. The second controller **100** is, for example, accommodated in the housing **30**. In an example, the second controller **100** includes a processor that executes a predetermined control program. The processor includes, for example, a CPU or an MPU. The processor can be provided at separate positions. The second controller **100** can include one or more microcomputers. The second controller **100** further includes, for example, an inverter circuit. The inverter circuit is electrically connected to the processor and the motor **98**. The second controller **100** can include a processor shared with the first controller **22A**. The second controller **100** can also be referred to as a motor controller, or an electronic motor controller, or a second electronic controller.

(207) The drive unit can further include a torque sensor configured to output a signal in accordance with the human driving force input to the crank axle **10**. The controller is, for example, configured to control the motor **98** in accordance with a detection result of a torque sensor. The torque sensor includes, for example, a strain gauge, a magnetostrictive sensor, or another known sensor. The torque sensor can be, for example, provided on the speed increaser **26**, the crank axle **10**, or the crank arms **12**.

(208) The drive unit includes, for example, second storage. The second storage stores various control programs and information used for various control processes. The storage includes, for example, a nonvolatile memory and a volatile memory. The nonvolatile memory includes, for example, at least one of a ROM, an EPROM, an EEPROM, and a flash memory. The volatile memory includes, for example, a RAM. The second storage can include a memory shared with the first storage. Thus, the second storage can be any computer storage device or any non-transitory computer-readable medium with the sole exception of a transitory, propagating signal.

(209) Modifications

(210) The description related to the above embodiments exemplifies, without any intention to limit, applicable forms of a component for a human-powered vehicle and a transmission device for a human-powered vehicle according to the present disclosure. The human-powered vehicle component and the human-powered vehicle transmission device according to the present disclosure can be applied to, for example, modifications of the embodiments that are described below and combinations of at least two of the modifications that do not contradict each other. In the following modifications, same reference characters are given to those components that are the same as the corresponding components of the above embodiments. Such components will not be described in detail.

(211) At least one of the minimum transmission ratio and the maximum transmission ratio of the transmission device **22** can be changed. The minimum transmission ratio of the transmission device

22 can be greater than one. The maximum transmission ratio of the transmission device **22** can be less than or equal to one. The transmission ratio of the transmission device **22** can include one. The minimum transmission ratio of the transmission device **22** can be one, and the maximum transmission ratio of the transmission device **22** can be 2.6.

(212) The number of transmission stages of the transmission device **22** can be changed. For example, in each embodiment, the first transmission unit can be omitted, and the second transmission unit can be omitted.

(213) The component **20** can be provided on the axle of the wheel of the human-powered vehicle. In this modified example, the component **20** is, for example, provided on the axle of the driving wheel. The driving wheel includes, for example, the rear wheel. The shaft member **24** includes, for example, a hub axle. The input rotational shaft **28** is, for example, not provided on the housing **30**. The output rotor **34** includes, for example, a hub shell. The output rotational portion **32** is, for example, formed integrally with the output rotor **34**. The hub shell is, for example, formed integrally with the housing **30**. The hub shell can be formed separately from the housing **30** and configured to rotate integrally with the housing **30**. The component **20** is provided, for example, on the rear hub of the rear wheel.

(214) The rotational torque of the motor **98** can be configured to be input to the input rotational portion **54**. In a case where the rotational torque of the motor **98** is input to the input rotational portion **54**, the reduction ratio of the motor speed reducer **98B** is further decreased. This allows for further reduction in size of the motor speed reducer **98B**.

(215) The motor speed reducer **98B** can be omitted. In a case where the motor speed reducer **98B** is omitted, for example, the rotational torque of the motor **98** is input to the input rotational portion **54** or the input rotational portion **54**.

(216) In this specification, the phrase “at least one of” as used in this disclosure means “one or more” of a desired choice. As one example, the phrase “at least one of” as used in this disclosure means “only one choice” or “both of two choices” in a case where the number of choices is two. In another example, in this specification, the phrase “at least one of” as used in this disclosure means “only one single choice” or “any combination of equal to or more than two choices” if the number of its choices is equal to or more than three.

(217) In this specification, ordinal numbers such as “first, second, and third” are used to merely distinguish between members having the same name and have no particular meaning.

Claims

1. A transmission device for a human-powered vehicle, the transmission device configured to switch a transmission stage of the human-powered vehicle in steps, the transmission device comprising: a shaft member; a first planetary gear unit provided on the shaft member; and a second planetary gear unit provided on the shaft member, the first planetary gear unit including at least one first planetary gear, a first carrier including at least one first carrier pin, the first carrier supporting the at least one first planetary gear with the at least one first carrier pin, at least one first sun gear meshed with the at least one first planetary gear, and at least one first ring gear meshed with the at least one first planetary gear; the second planetary gear unit includes at least one second planetary gear, a second carrier including at least one second carrier pin, the second carrier supporting the at least one second planetary gear with the at least one second carrier pin, at least one second sun gear meshed with the at least one second planetary gear, and at least one second ring gear meshed with the at least one second planetary gear; the at least one first carrier pin being a member separate from the at least one second carrier pin and arranged at a distance from the at least one second carrier pin; the at least one first carrier pin and the at least one second carrier pin being arranged overlapping a first plane that extends perpendicular to the shaft member; and a rotational state of at least one of the at least one first sun gear, the at least one second sun gear, the at least one first ring

gear, and the at least one second ring gear being controlled to switch the transmission stage in steps.

2. The transmission device according to claim 1, wherein: the at least one first planetary gear includes a plurality of first planetary gears of different diameters; and at least one of the first planetary gears having a maximum diameter is arranged overlapping the at least one second planetary gear as viewed in an axial direction of the shaft member.

3. The transmission device according to claim 1, wherein: the at least one second planetary gear includes a plurality of second planetary gears of different diameters; and at least one of the second planetary gears having a maximum diameter is arranged overlapping the at least one first planetary gear as viewed in an axial direction of the shaft member.

4. The transmission device according to claim 1, further comprising: a third planetary gear unit provided on the shaft member, the third planetary gear unit including at least one third planetary gear, a third carrier including at least one third carrier pin, the third carrier supporting the at least one third planetary gear with the at least one third carrier pin, at least one third sun gear meshed with the at least one third planetary gear, and at least one third ring gear meshed with the at least one third planetary gear; and a rotational state of at least one of the at least one third sun gear and the at least one third ring gear being controlled to switch the transmission stage in steps.

5. The transmission device according to claim 1, wherein a part of the first carrier is formed integrally with a part of the second carrier.

6. The transmission device according to claim 5, wherein: the first carrier includes a first carrier portion and a second carrier portion; the at least one first carrier pin is supported by the first carrier portion and the second carrier portion; and the at least one first planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member.

7. The transmission device according to claim 6, wherein: the second carrier includes the first carrier portion and the second carrier portion; the at least one second carrier pin is supported by the first carrier portion and the second carrier portion; and the at least one second planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member.

8. The transmission device according to claim 4, wherein: the first carrier includes a first carrier portion and a second carrier portion; the at least one first carrier pin is supported by the first carrier portion and the second carrier portion; the at least one first planetary gear is located between the first carrier portion and the second carrier portion in an axial direction of the shaft member; the second carrier includes a third carrier portion and at least one of the first carrier portion and the second carrier portion; the at least one second carrier pin is supported by the third carrier portion and the at least one of the first carrier portion and the second carrier portion; the at least one second planetary gear is located between the first carrier portion and the third carrier portion; and the second carrier is located between the first carrier portion and the third carrier portion in an axial direction of the shaft member.

9. The transmission device according to claim 1, further comprising: a transmission switching mechanism including a plurality of clutches, the transmission switching mechanism being configured to switch the rotational state of the at least one first ring gear and the rotational state of the at least one second ring gear to switch the transmission stage in steps.

10. The transmission device according to claim 9, wherein: the at least one first planetary gear includes first planetary gears; the at least one first ring gear includes first ring gears that are rotatable independently from each other; the first planetary gears are each meshed with the first ring gears; and the transmission switching mechanism is configured to switch rotational states of the first ring gears to switch the transmission stage in steps.

11. The transmission device according to claim 9, wherein: the at least one second planetary gear includes second planetary gears; the at least one second ring gear includes second ring gears that are rotatable independently from each other; the second planetary gears are each meshed with the

second ring gears; and the transmission switching mechanism is configured to switch rotational states of the second ring gears to switch the transmission stage in steps.

12. The transmission device according to claim 9, wherein the at least one first sun gear and the at least one second sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

13. The transmission device according to claim 1, wherein: the at least one first planetary gear includes first planetary gears arranged at intervals about a rotational center axis of the shaft member; the at least one second planetary gear includes second planetary gears arranged at intervals about the rotational center axis of the shaft member; the at least one first carrier pin includes first carrier pins corresponding to the first planetary gears, the at least one second carrier pin includes second carrier pins corresponding to the second planetary gears; and the first carrier pins and the second carrier pins are arranged alternately at equal intervals about the rotational center axis of the shaft member.

14. The transmission device according to claim 1, further comprising: a third planetary gear unit provided on the shaft member; and a fourth planetary gear unit provided on the shaft member, the third planetary gear unit including at least one third planetary gear, a third carrier including at least one third carrier pin, the third carrier supporting the at least one third planetary gear with the at least one third carrier pin, at least one third sun gear meshed with the at least one third planetary gear, and at least one third ring gear meshed with the at least one third planetary gear; the fourth planetary gear unit including at least one fourth planetary gear, a fourth carrier including at least one fourth carrier pin, the fourth carrier supporting the at least one fourth planetary gear with the at least one fourth carrier pin, at least one fourth sun gear meshed with the at least one fourth planetary gear, and at least one fourth ring gear meshed with the at least one fourth planetary gear; the at least one third carrier pin is a member separate from the at least one fourth carrier pin and arranged at a distance from the at least one fourth carrier pin; the at least one third carrier pin and the at least one fourth carrier pin are arranged overlapping a second plane that extends perpendicular to the shaft member; and a rotational state of at least one of the at least one third sun gear, the at least one fourth sun gear, the at least one third ring gear, and the at least one fourth ring gear is controlled to switch the transmission stage in steps.

15. The transmission device according to claim 14, wherein a part of the third carrier is formed integrally with a part of the fourth carrier.

16. The transmission device according to claim 14, wherein the at least one third sun gear and the at least one fourth sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

17. The transmission device according to claim 14, further comprising: an input rotational portion; and an output rotational portion, the input rotational portion being connected to the first carrier, and the output rotational portion being connected to the fourth carrier.

18. The transmission device according to claim 14, wherein the at least one first sun gear, the at least one second sun gear, the at least one third sun gear, and the at least one fourth sun gear are configured to rotate integrally in at least a first direction about a rotational center axis of the shaft member.

19. The transmission device according to claim 1, further comprising: a housing attachable to a frame of the human-powered vehicle, the transmission device being provided in the housing and configured to be connected to a rear wheel of the human-powered vehicle by at least one of a chain, a belt, and a shaft in a state in which the housing is attached to the frame.

20. The transmission device according to claim 1, further comprising: an input rotational shaft, the input rotational shaft being arranged parallel to the shaft member and distanced from the shaft member in a radial direction of the input rotational shaft.

21. A component for a human-powered vehicle, the component comprising: the transmission device according to claim 1; and a motor configured to apply a propulsion force to the human-powered

vehicle.

22. The transmission device according to claim 9, wherein the transmission switching mechanism further includes an electric actuator.
